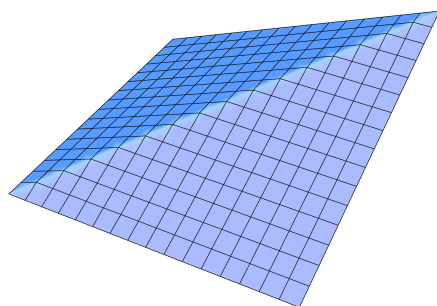




STAT 433: Stochastic Processes

Prof. Yi Shen • Fall 2014 • University of Waterloo
MW 2:30–3:50PM in MC 4061

Transcribed, $\text{\LaTeX}$ ed, and edited by Rob Zimmerman

Note: These are personal lecture notes, not official course materials. They may contain errors (which by default you should attribute to me, Rob Zimmerman, rather than the course instructor). The permanent home of these — and all of my other lecture notes — is <https://rob-zimmerman.github.io/coursenotes>.

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Lecture 1 — Monday, September 08, 2014

1. Discrete-Time Markov Chains

Definitions and Transition Matrices

We start off by reviewing some familiar concepts. A **random variable** is a real-valued function of the outcomes of a random experiment. Thus $X : \Omega \rightarrow \mathbb{R}$, where $\Omega = \{\omega_1, \omega_2, \dots\}$ is the sample space. A **stochastic process** is a family of random variables $\{X_t\}_{t \in T}$ defined on a common sample space Ω . The index set T often represents time, although this need not always be the

interpretation. In discrete time, the usual index set is $T = \{0, 1, 2, \dots\}$ and we write X_n for $n = 0, 1, 2, \dots$, so that

$$\{X_n : n = 0, 1, 2, \dots\} = \{X_n\}_{n=0,1,2,\dots}.$$

In continuous time, one often uses $T = \{t \mid t \geq 0\}$ and writes

$$\{X_t : t \geq 0\} = \{X_t\}_{t \geq 0}.$$

The **state space** S is the collection of all possible values of the random variables in the process.

Definition 1.1 A stochastic process $\{X_n : n = 0, 1, 2, \dots\}$ is called a **discrete-time Markov chain** if the following hold:

1. The state space S is at most countable, meaning that it is either finite or countably infinite.
2. It satisfies the **Markov property**: for any $n = 0, 1, 2, \dots$,

$$\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, X_{n-1} = x_{n-1}, \dots) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

In other words, the future depends only on the present state.

Definition 1.2 The **transition probability** from state $i \in S$ at time n to state $j \in S$ at time $n + 1$ is given by

$$P_{n,i,j} = \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

for $n = 0, 1, 2, \dots$. In this course, we focus on the case where $P_{n,i,j}$ does not depend on n , and simply write $P_{i,j}$. In this case, the Markov chain is called **time-homogeneous**, or simply **homogeneous**. This will be the default discrete-time Markov chain setting. The matrix $\mathbf{P} = \{P_{i,j}\}_{i,j \in S}$ is called the **one-step transition probability matrix**.

The transition matrix satisfies

$$P_{i,j} \geq 0 \quad \text{for all } i, j \in S$$

and

$$\sum_{j \in S} P_{i,j} = 1,$$

so the entries in each row add to 1. The n -step transition probability is

$$P_{i,j}^{(n)} = \mathbb{P}(X_{n+m} = j \mid X_m = i) = \mathbb{P}(X_n = j \mid X_0 = i)$$

and the corresponding matrix is $\mathbf{P}^{(n)} = \{P_{i,j}^{(n)}\}_{i,j \in S}$. The one-step and n -step transition matrices are related by the Chapman-Kolmogorov equation.

Theorem 1.3 $\mathbf{P}^{(n)} = \mathbf{P}^n$.

Proof. We prove the result by induction. The case $n = 1$ follows from the definition, and $\mathbf{P}^{(0)} = \mathbf{I} = \mathbf{P}^0$. Assume that $\mathbf{P}^{(n)} = \mathbf{P}^n$. Then

$$\begin{aligned}
 P_{i,j}^{(n+1)} &= \mathbb{P}(X_{n+1} = j \mid X_0 = i) \\
 &= \sum_{k \in S} \mathbb{P}(X_{n+1} = j, X_n = k \mid X_0 = i) \\
 &= \sum_{k \in S} \mathbb{P}(X_{n+1} = j \mid X_n = k, X_0 = i) \mathbb{P}(X_n = k \mid X_0 = i) \\
 &= \sum_{k \in S} \mathbb{P}(X_{n+1} = j \mid X_n = k) \mathbb{P}(X_n = k \mid X_0 = i) \\
 &= \sum_{k \in S} P_{k,j} P_{i,k}^{(n)} \\
 &= (\mathbf{P}^{(n)} \mathbf{P})_{i,j} \\
 &= (\mathbf{P}^n \mathbf{P})_{i,j} = (\mathbf{P}^{n+1})_{i,j}.
 \end{aligned}$$

□

In particular, this implies that $\mathbf{P}^{(n)} = \mathbf{P}^{(m)} \cdot \mathbf{P}^{(n-m)}$ for $0 \leq m \leq n$, or equivalently

$$P_{i,j}^{(n)} = \sum_{k \in S} P_{i,k}^{(m)} P_{k,j}^{(n-m)} \quad \text{for all } i, j \in S, \ 0 \leq m \leq n.$$

This is the **Chapman-Kolmogorov equation**.

Let $\boldsymbol{\alpha}_n = (\alpha_{n,0}, \alpha_{n,1}, \dots)^\top$ be the distribution of X_n , where $\alpha_{n,k} = \mathbb{P}(X_n = k)$ for all k . Then $\alpha_{n,k} \geq 0$ for all n, k , and

$$\sum_{k \in S} \alpha_{n,k} = 1$$

for all $n = 0, 1, 2, \dots$. The distribution

$$\boldsymbol{\alpha}_0 = (\mathbb{P}(X_0 = 0), \mathbb{P}(X_0 = 1), \mathbb{P}(X_0 = 2), \dots)^\top$$

is called the initial distribution.

Theorem 1.4 $\alpha_n = (\mathbf{P}^\top)^n \alpha_0$.

Proof. For $j \in S$,

$$\alpha_{n,j} = \mathbb{P}(X_n = j) = \sum_{i \in S} \mathbb{P}(X_n = j \mid X_0 = i) \mathbb{P}(X_0 = i) = \sum_{i \in S} ((\mathbf{P}^\top)^n)_{j,i} \alpha_{0,i} = \left((\mathbf{P}^\top)^n \alpha_0 \right)_j.$$

□

Lecture 2 — Wednesday, September 10, 2014

The probabilistic behaviour of a homogeneous discrete-time Markov chain is determined by its initial distribution and its transition probability matrix. We saw that $\mathbf{P}^{(n)} = \mathbf{P}^n$, and so $\mathbf{P}^{(n)} = \mathbf{P}^{(m)} \mathbf{P}^{(n-m)}$.

More generally, for any $n \in \mathbb{N}$, the finite-dimensional distribution is

$$\begin{aligned} \mathbb{P}(X_n = x_n, \dots, X_0 = x_0) &= \mathbb{P}(X_0 = x_0) \mathbb{P}(X_1 = x_1 \mid X_0 = x_0) \mathbb{P}(X_2 = x_2 \mid X_1 = x_1) \\ &\quad \dots \mathbb{P}(X_n = x_n \mid X_{n-1} = x_{n-1}) \\ &= \alpha_{0,x_0} P_{x_0,x_1} P_{x_1,x_2} \dots P_{x_{n-1},x_n} \end{aligned}$$

by repeatedly applying the Markov property. Even more generally, if

$$0 \leq t_1 < \dots < t_n,$$

then

$$\mathbb{P}(X_{t_n} = x_{t_n}, \dots, X_{t_1} = x_{t_1}) = \mathbb{P}(X_{t_1} = x_{t_1}) (\mathbf{P}^{t_2-t_1})_{x_{t_1},x_{t_2}} \dots (\mathbf{P}^{t_n-t_{n-1}})_{x_{t_{n-1}},x_{t_n}},$$

where

$$\mathbb{P}(X_{t_1} = x_{t_1}) = \alpha_{0,x_{t_1}} = (\alpha_0 \cdot \mathbf{P}^{t_1})_{x_{t_1}} = \sum_{k \in S} \alpha_{0,k} (\mathbf{P}^{t_1})_{k,x_{t_1}}.$$

This again shows that the probabilistic properties of a homogeneous discrete-time Markov chain are fully characterized by the initial distribution and the transition matrix $\mathbf{P}$.

Classification of States

Definition 2.1 State j is **accessible** from state i if there exists $n \in \{0, 1, 2, \dots\}$ such that $P_{i,j}^{(n)} > 0$. Otherwise, j is not accessible from i . If, in addition, i is accessible from j , then i and j **communicate**, and in this case we write $i \leftrightarrow j$.

Intuitively, if state j is accessible from state i , then it is possible to reach j from i in a finite number of steps.

The communication relation $\leftrightarrow$ is an equivalence relation. It is reflexive since $P_{i,i}^{(0)} = 1 > 0$, and it is symmetric by definition. It is also transitive: if $i \leftrightarrow j$ and $j \leftrightarrow k$, then there exist m, n such that

$$P_{i,k}^{(m+n)} = \sum_{l \in S} P_{i,l}^{(m)} P_{l,k}^{(n)} \geq P_{i,j}^{(m)} P_{j,k}^{(n)} > 0.$$

Thus i is accessible from k , and the reverse accessibility follows similarly. Therefore $i \leftrightarrow k$. The relation $\leftrightarrow$ partitions the state space into disjoint equivalence classes: states that communicate with each other all belong to one class. These classes form a partition of S , meaning that $S_i \cap S_j = \emptyset$ for $i \neq j$ and $\bigcup_i S_i = S$.

One can find the equivalence classes by drawing a directed graph whose nodes are the states and whose directed edges are $i \rightarrow j$ whenever $P_{i,j} > 0$.

Definition 2.2 A Markov chain with only one equivalence class is called **irreducible**.

Definition 2.3 The number

$$d_{(i)} := \gcd\{n \geq 1 \mid P_{i,i}^{(n)} > 0\}$$

is called the **period** of state i . If $P_{i,i}^{(n)} = 0$ for all $n > 0$, we set $d_{(i)} = \infty$. The state is called **aperiodic** if $d_{(i)} = 1$. The Markov chain is called **aperiodic** if all of its states are aperiodic.

Theorem 2.4 If $i \leftrightarrow j$, then $d_{(i)} = d_{(j)}$.

Proof. Since $i \leftrightarrow j$, there exist m, n such that $P_{i,j}^{(m)} > 0$ and $P_{j,i}^{(n)} > 0$. For any l such that

$$P_{j,j}^{(l)} > 0,$$

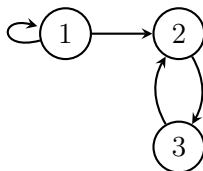
$$P_{i,i}^{(m+l+n)} \geq P_{i,j}^{(m)} P_{j,j}^{(l)} P_{j,i}^{(n)} > 0.$$

On the other hand,

$$P_{i,i}^{(m+n)} \geq P_{i,j}^{(m)} P_{j,i}^{(n)} > 0.$$

Since $d_{(i)}$ divides both $m+n$ and $m+n+l$, it divides l . Therefore $d_{(i)}$ divides $\gcd\{l \mid P_{j,j}^{(l)} > 0\} = d_{(j)}$. By symmetry, $d_{(j)}$ divides $d_{(i)}$, and hence $d_{(i)} = d_{(j)}$. $\square$

Consider the Markov chain with the following directed graph representation:



State 1 has a self-loop, while states 2 and 3 communicate with each other. Note that $d_{(1)} = 1$ even if the one-step probability $P_{1,1}$ is 0. We thus make the following remarks:

Remark 2.5

1. In general, $d_{(i)} = k \not\Rightarrow P_{i,i}^{(k)} > 0$.
2. If the Markov chain is irreducible, all states have the same period. In this case, we also call it the period of the Markov chain.

Recurrence and Transience

For $n \in \mathbb{Z}^+$, define

$$f_{i,j}^{(n)} := \mathbb{P}(X_n = j, X_{n-1} \neq j, \dots, X_1 \neq j \mid X_0 = i).$$

Intuitively, this is the probability that the chain visits state j for the first time at time n , given that it starts from state i .

The quantities $P_{i,j}^{(n)}$ and $f_{i,j}^{(n)}$ are related as follows. First, $P_{i,j}^{(n)} \geq f_{i,j}^{(n)}$. Also,

$$\sum_{n=1}^{\infty} f_{i,j}^{(n)} = \mathbb{P}(\text{the Markov chain ever visits } j \mid X_0 = i) \leq 1.$$

More precisely,

$$P_{i,j}^{(n)} = \sum_{k=1}^n f_{i,j}^{(k)} P_{j,j}^{(n-k)} = f_{i,j}^{(n)} + \sum_{k=1}^{n-1} f_{i,j}^{(k)} P_{j,j}^{(n-k)}.$$

Lecture 3 — Monday, September 15, 2014

Definition 3.1 Let

$$\begin{aligned} f_{i,j}^{(n)} &:= \mathbb{P}(X_n = j, X_{n-1} \neq j, \dots, X_1 \neq j \mid X_0 = i) \\ &= \mathbb{P}(\text{the Markov chain visits } j \text{ first on the } n\text{-th step} \mid X_0 = i). \end{aligned}$$

We know that $P_{i,j}^{(n)} \geq f_{i,j}^{(n)}$ and

$$\sum_{n=1}^{\infty} f_{i,j}^{(n)} \leq 1.$$

The relation between $P_{i,j}^{(n)}$ and $f_{i,j}^{(n)}$ is this:

Theorem 3.2

$$P_{i,j}^{(n)} = \sum_{k=1}^n f_{i,j}^{(k)} P_{j,j}^{(n-k)} = f_{i,j}^{(n)} + \sum_{k=1}^{n-1} f_{i,j}^{(k)} P_{j,j}^{(n-k)}.$$

Proof.

$$\begin{aligned} P_{i,j}^{(n)} &= \mathbb{P}(X_n = j \mid X_0 = i) \\ &= \sum_{k=1}^n \mathbb{P}(X_n = j, X \text{ first visits } j \text{ at time } k \mid X_0 = i) \\ &= \sum_{k=1}^n \mathbb{P}(X_n = j \mid X \text{ first visits } j \text{ at time } k, X_0 = i) \mathbb{P}(X \text{ first visits } j \text{ at time } k \mid X_0 = i) \\ &= \sum_{k=1}^n \mathbb{P}(X_n = j \mid X_k = j) f_{i,j}^{(k)} \\ &= \sum_{k=1}^n f_{i,j}^{(k)} P_{j,j}^{(n-k)} \end{aligned}$$

$$= f_{i,j}^{(n)} + \sum_{k=1}^{n-1} f_{i,j}^{(k)} P_{j,j}^{(n-k)}.$$

□

We can rewrite this as

$$f_{i,j}^{(n)} = P_{i,j}^{(n)} - \sum_{k=1}^{n-1} f_{i,j}^{(k)} P_{j,j}^{(n-k)},$$

which is a recursive formula to calculate $f_{i,j}^{(n)}$.

Definition 3.3 We define

$$f_{i,j} := \mathbb{P}(\text{the Markov chain ever goes to state } j \mid X_0 = i) = \sum_{n=1}^{\infty} f_{i,j}^{(n)}.$$

Definition 3.4 State i is called **transient** if $f_{i,i} < 1$. Otherwise, state i is called **recurrent** (i.e., when $f_{i,i} = 1$).

Another way to characterize recurrence and transience is as follows. Define

$$M_i = \text{total number of times the Markov chain visits state } i \text{ after time } 0 = \sum_{n=1}^{\infty} \mathbb{1}_{\{X_n=i\}}.$$

If $f_{i,i} < 1$, then

$$\mathbb{P}(M_i = k \mid X_0 = i) = f_{i,i}^{k-1} (1 - f_{i,i}).$$

Thus, given $X_0 = i$, M_i follows a geometric distribution with parameter $1 - f_{i,i}$, and

$$\mathbb{E}[M_i \mid X_0 = i] = \frac{f_{i,i}}{1 - f_{i,i}} < \infty$$

when $f_{i,i} < 1$. On the other hand, notice that

$$\lim_{f_{i,i} \rightarrow 1} \frac{f_{i,i}}{1 - f_{i,i}} = \infty.$$

Alternatively,

$$\mathbb{E}[M_i \mid X_0 = i] = \sum_{n=1}^{\infty} \mathbb{P}(X_n = i \mid X_0 = i) = \sum_{n=1}^{\infty} P_{i,i}^{(n)}.$$

If $f_{i,i} = 1$, then $\mathbb{E}[M_i \mid X_0 = i] = \infty$. Therefore i is recurrent if and only if $\mathbb{E}[M_i \mid X_0 = i] = \infty$, and i is transient if and only if $\mathbb{E}[M_i \mid X_0 = i] < \infty$.

Remark 3.5 We actually have recurrence if and only if $M_i = \infty$ almost surely, and transience if and only if $M_i < \infty$ almost surely.

Yet another way to see the situation is to observe that

$$\begin{aligned} \mathbb{E}[M_i \mid X_0 = i] &= \mathbb{E}\left[\sum_{n=1}^{\infty} \mathbb{1}_{\{X_n=i\}} \mid X_0 = i\right] \\ &= \sum_{n=1}^{\infty} \mathbb{E}[\mathbb{1}_{\{X_n=i\}} \mid X_0 = i] \\ &= \sum_{n=1}^{\infty} \mathbb{P}(X_n = i \mid X_0 = i) \\ &= \sum_{n=1}^{\infty} P_{i,i}^{(n)}. \end{aligned}$$

Thus i is recurrent if and only if

$$\sum_{n=1}^{\infty} P_{i,i}^{(n)} = \infty.$$

Recurrence and transience are class properties.

Theorem 3.6 If $i \leftrightarrow j$ and i is recurrent, then j is also recurrent.

Proof. Since $i \leftrightarrow j$, there exist m, n such that $P_{i,j}^{(m)} > 0$ and $P_{j,i}^{(n)} > 0$. Then

$$\sum_{l=1}^{\infty} P_{j,j}^{(l)} \geq \sum_{l=1}^{\infty} P_{j,i}^{(n)} P_{i,i}^{(l)} P_{i,j}^{(m)} = P_{j,i}^{(n)} P_{i,j}^{(m)} \sum_{l=1}^{\infty} P_{i,i}^{(l)}.$$

Since i is recurrent,

$$\sum_{l=1}^{\infty} P_{i,i}^{(l)} = \infty.$$

Thus

$$\sum_{l=1}^{\infty} P_{j,j}^{(l)} = \infty,$$

so j is recurrent. □

Corollary 3.7 If $i \leftrightarrow j$ and i is transient, then j is transient as well.

Consequently, if a Markov chain is irreducible, then either all states are transient or all states are recurrent. The transient case is impossible when the state space is finite.

Theorem 3.8 If i is recurrent and i does not communicate with state j , then $P_{i,j} = 0$.

Proof. Suppose for a contradiction that $P_{i,j} > 0$. Since i and j do not communicate, i is not accessible from j . Then

$$\begin{aligned} f_{i,i} &= \mathbb{P}(\text{the Markov chain visits } i \text{ again} \mid X_0 = i) \\ &= 1 - \mathbb{P}(\text{the Markov chain never visits } i \text{ again} \mid X_0 = i) \\ &\leq 1 - P_{i,j} \\ &< 1. \end{aligned}$$

This inequality holds because $\{X_1 = j\} \subseteq \{\text{the Markov chain never visits } i \text{ again}\}$. This contradicts the recurrence of i . Hence $P_{i,j} = 0$. □

Remark 3.9 Under the assumptions of the preceding theorem, one can in fact show that j is not accessible from i ; that is, $P_{i,j}^{(n)} = 0$ for every $n = 1, 2, \dots$. The key point is that if j were accessible from i , then there would exist some $n > 0$ such that

$$\mathbb{P}(X_n = j, X_{n-1} \neq j, \dots, X_1 \neq j \mid X_0 = i) > 0.$$

This would contradict the recurrence of i .

Corollary 3.10 If j is accessible from i but i is not accessible from j , then i must be transient.

Lecture 4 — Wednesday, September 17, 2014

Positive Recurrence and Stationary Distributions

Recall that state i is recurrent when $f_{i,i} = 1$ and transient when $f_{i,i} < 1$. Equivalently, i is recurrent when $\mathbb{E}[M_i \mid X_0 = i] = \infty$, where

$$M_i = \sum_{n=1}^{\infty} \mathbb{1}_{\{X_n=i\}},$$

the number of visits to i after time 0. Thus i is transient if and only if $\mathbb{E}[M_i \mid X_0 = i] < \infty$, while i is recurrent if and only if

$$\sum_{n=1}^{\infty} P_{i,i}^{(n)} = \infty.$$

Similarly, i is transient if and only if

$$\sum_{n=1}^{\infty} P_{i,i}^{(n)} < \infty.$$

If $i \leftrightarrow j$ and i is recurrent, then j is recurrent. Hence transience and recurrence are class properties. Also, if i is recurrent and $i \not\rightarrow j$, then $P_{i,j} = 0$. Consequently, if j is accessible from i but i is not accessible from j , then i is transient.

Remark 4.1 This shows that once a Markov chain enters a recurrent class, it never leaves that class.

Definition 4.2 Define

$$R_i := \min\{n \geq 1 \mid X_n = i\}$$

to be the first return time to state i .

We have

$$\mathbb{P}(R_i = k \mid X_0 = i) = f_{i,i}^{(k)}.$$

If i is recurrent, then

$$\sum_{k=1}^{\infty} f_{i,i}^{(k)} = f_{i,i} = 1$$

and

$$\mathbb{P}(R_i = \infty \mid X_0 = i) = 0.$$

However, recurrence does *not* imply that

$$\mathbb{E}[R_i \mid X_0 = i] = \sum_{k=1}^{\infty} k \cdot f_{i,i}^{(k)} < \infty.$$

Definition 4.3 Let i be recurrent. The state i is called **positive recurrent** if

$$\mathbb{E}[R_i \mid X_0 = i] < \infty,$$

and **null recurrent** if

$$\mathbb{E}[R_i \mid X_0 = i] = \infty.$$

Positive and null recurrence are class properties.

Theorem 4.4 Let $i \leftrightarrow j$ and suppose i is positive recurrent. Then j is also positive recurrent.

The proof is beyond the scope of our course.

Definition 4.5 A discrete-time Markov chain is called **ergodic** if it is irreducible, positive recurrent, and aperiodic.

Stationary Distributions

Definition 4.6 A sequence $\{p_i\}_{i \geq 0}$ with $p_i \geq 0$ for all i is called a **stationary distribution** for a discrete-time Markov chain with transition matrix $\mathbf{P}$ if

$$\sum_{i=0}^{\infty} p_i = 1$$

and

$$p_j = \sum_{i=0}^{\infty} p_i P_{i,j} \quad \text{for all } j.$$

If $\mathbf{p} = (p_0, p_1, \dots)^\top$, then these conditions are $\mathbf{p} = \mathbf{P}^\top \mathbf{p}$ and $\mathbf{1}_{|S|}^\top \mathbf{p} = 1$.

This distribution is called stationary because if the chain has initial distribution $\alpha_0 = p$, then

$$\alpha_n = (\mathbf{P}^\top)^n \alpha_0 = (\mathbf{P}^\top)^n \mathbf{p} = \mathbf{p}.$$

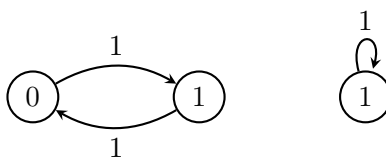
Thus the (marginal) distribution of the chain does not change over time. The word “marginal” can be omitted here since the distribution of a homogeneous Markov chain is completely determined by its initial distribution and transition matrix.

Question 4.7 Does a stationary distribution always exist? And when it does exist, is it unique?

In general, the answer to both questions is no. A stationary distribution does not exist if all states of the chain are either null recurrent or transient. It is not unique if the chain has more than one positive recurrent equivalence class.

Example 4.8 Let

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$



This chain has two closed classes: $\{1\}$ and $\{0, 2\}$. Two stationary distributions are

$$\mathbf{p}^{(1)} = (0, 1, 0)^\top, \quad \mathbf{p}^{(2)} = \left(\frac{1}{2}, 0, \frac{1}{2}\right)^\top.$$

Hence stationary distributions are not unique. Any convex combination

$$\alpha \mathbf{p}^{(1)} + (1 - \alpha) \mathbf{p}^{(2)} = \left(\frac{1 - \alpha}{2}, \alpha, \frac{1 - \alpha}{2} \right), \quad \alpha \in [0, 1],$$

is also stationary.

A stationary distribution is sometimes called an **invariant probability**. For a finite state space, the stationary distributions are found by solving the system $\mathbf{p} = \mathbf{P}^\top \mathbf{p}$ and $\mathbf{1}_N^\top \mathbf{p} = 1$. With N states, we have

$$p_j = \sum_{i=0}^{N-1} p_i P_{i,j}$$

for $j = 0, 1, \dots, N - 1$, and

$$\sum_{j=0}^{N-1} p_j = 1.$$

There are N unknowns $(p_0, \dots, p_{N-1})$ and $N + 1$ equations. However, if we add the first N equations, we obtain

$$\sum_{j=0}^{N-1} p_j = \sum_{j=0}^{N-1} \sum_{i=0}^{N-1} p_i P_{i,j} = \sum_{i=0}^{N-1} p_i \sum_{j=0}^{N-1} P_{i,j} = \sum_{i=0}^{N-1} p_i.$$

Thus we have linear dependence, so we can drop one of the $N + 1$ equations, and there exists a non-zero solution.

Limiting Distributions

We are interested in $\lim_{n \rightarrow \infty} P_{i,j}^{(n)}$ and $\lim_{n \rightarrow \infty} \alpha_n$. Periodicity and recurrence play an important role in the limiting behaviour of a Markov chain.

Example 4.9 Using the same 3-state Markov chain as in [Example 4.8](#),

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Since

$$\mathbf{P}^2 = \mathbf{I},$$

we have

$$\mathbf{P}^{2n} = \mathbf{I}, \quad \mathbf{P}^{2n+1} = \mathbf{P}.$$

Thus $\mathbf{P}^{2n+1} \neq \mathbf{P}^{2n}$, so the limit of $\mathbf{P}^n$ does not exist.

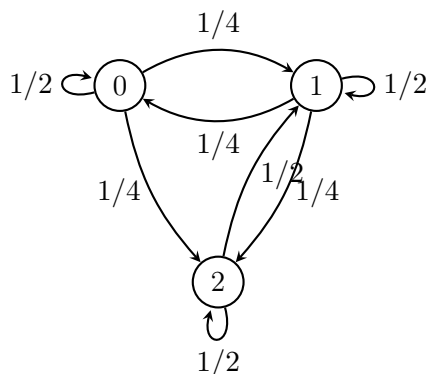
Lecture 5 — Monday, September 22, 2014

Examples show that, in general, a stationary distribution need not exist, and when it exists it need not be unique.

We are interested in limits such as $\lim_{n \rightarrow \infty} P_{i,j}^{(n)}$, $\lim_{n \rightarrow \infty} \alpha_n$, and $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = j)$.

Example 5.1 Let

$$\mathbf{P} = \begin{pmatrix} 2/4 & 1/4 & 1/4 \\ 1/4 & 2/4 & 1/4 \\ 0 & 1/2 & 1/2 \end{pmatrix}.$$



This chain has one class, so it is irreducible and recurrent. Moreover, $d(i) = 1$ for $i = 0, 1, 2$.

For this example,

$$\lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} 4/11 & 4/11 & 3/11 \\ 4/11 & 4/11 & 3/11 \\ 4/11 & 4/11 & 3/11 \end{pmatrix}.$$

This limit can be found numerically, up to round-off error, or by hand using diagonalization.

If $\mathbf{P} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1}$, where $\mathbf{Q}$ is invertible and $\mathbf{D}$ is diagonal, then

$$\begin{aligned}\lim_{n \rightarrow \infty} \mathbf{P}^n &= \lim_{n \rightarrow \infty} (\mathbf{Q}\mathbf{D}\mathbf{Q}^{-1})^n \\ &= \lim_{n \rightarrow \infty} \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1}\mathbf{Q}\mathbf{D}\mathbf{Q}^{-1} \dots \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1} \\ &= \mathbf{Q}(\lim_{n \rightarrow \infty} \mathbf{D}^n)\mathbf{Q}^{-1}.\end{aligned}$$

For the matrix $\mathbf{P}$, the eigenvalues are 1, 1/4, 1/4, so we can write

$$\mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}, \quad \mathbf{D}^n = \begin{pmatrix} 1 & 0 & 0 \\ 0 & (1/4)^n & 0 \\ 0 & 0 & (1/4)^n \end{pmatrix} \xrightarrow{n \rightarrow \infty} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence

$$\lim_{n \rightarrow \infty} \mathbf{P}^n = \mathbf{Q} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{Q}^{-1}.$$

If we write $\mathbf{Q} = [\mathbf{q}_1 \ \mathbf{q}_2 \ \mathbf{q}_3]$ and let $\mathbf{r}_1$ be the first row of $\mathbf{Q}^{-1}$, then

$$= \mathbf{Q} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{Q}^{-1} = \mathbf{q}_1 \mathbf{r}_1.$$

Since $\mathbf{P}$ is stochastic, $\mathbf{q}_1 = (1, 1, 1)^\top$. Taking

$$\mathbf{r}_1 = \left(\frac{4}{11}, \frac{4}{11}, \frac{3}{11} \right),$$

we get

$$\mathbf{q}_1 \mathbf{r}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 4/11 & 4/11 & 3/11 \end{pmatrix} = \begin{pmatrix} 4/11 & 4/11 & 3/11 \\ 4/11 & 4/11 & 3/11 \\ 4/11 & 4/11 & 3/11 \end{pmatrix}.$$

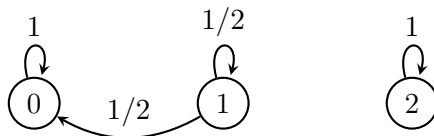
The limit of $\mathbf{P}^n$ exists, and the limiting matrix has identical rows. This implies that

$$\lim_{n \rightarrow \infty} \boldsymbol{\alpha}_n = \left(\lim_{n \rightarrow \infty} (\mathbf{P}^\top)^n \right) \boldsymbol{\alpha}_0 = (4/11, 4/11, 3/11)^\top.$$

Thus the limiting distribution of X_n exists and does not depend on the initial distribution.

Example 5.2 Let

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$



The states are $\{0\}, \{1\}, \{2\}$. The states $\{0\}$ and $\{2\}$ are recurrent, while $\{1\}$ is transient. The chain is aperiodic. Since $P_{00} = P_{22} = 1$, states 0 and 2 are absorbing. We have

$$\lim_{n \rightarrow \infty} P_{00}^{(n)} = \lim_{n \rightarrow \infty} P_{22}^{(n)} = 1,$$

while $\lim_{n \rightarrow \infty} P_{01}^{(n)} = \lim_{n \rightarrow \infty} P_{02}^{(n)} = \lim_{n \rightarrow \infty} P_{20}^{(n)} = \lim_{n \rightarrow \infty} P_{21}^{(n)} = 0$ and $P_{11}^{(n)} \rightarrow 0$. Thus

$$\lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} 1 & 0 & 0 \\ 2/3 & 0 & 1/3 \\ 0 & 0 & 1 \end{pmatrix}.$$

The limit of $\mathbf{P}^n$ exists, but the rows are not identical. As a result, the limiting distribution depends on the initial state. The column corresponding to state 1 is the zero vector in the limit, so $((\mathbf{P}^\top)^n \boldsymbol{\alpha}_0)_1 \rightarrow 0$ as $n \rightarrow \infty$ for every initial distribution $\boldsymbol{\alpha}_0$. Thus, no matter what the initial state or distribution is, the limiting distribution has no mass at state 1.

Proposition 5.3 Let j be a transient state of a Markov chain. Then for any state i ,

$$\lim_{n \rightarrow \infty} P_{i,j}^{(n)} = 0.$$

Proof.

$$\begin{aligned} \sum_{n=1}^{\infty} P_{i,j}^{(n)} &= \sum_{n=1}^{\infty} \sum_{k=1}^n f_{i,j}^{(k)} P_{j,j}^{(n-k)} \\ &= \sum_{k=1}^{\infty} f_{i,j}^{(k)} \sum_{n=k}^{\infty} P_{j,j}^{(n-k)} \end{aligned}$$

$$\begin{aligned}
&= \left(\sum_{k=1}^{\infty} f_{i,j}^{(k)} \right) \left(\sum_{m=0}^{\infty} P_{j,j}^{(m)} \right) \\
&= f_{i,j} \left(1 + \sum_{m=1}^{\infty} P_{j,j}^{(m)} \right) < \infty,
\end{aligned}$$

since j is transient. Therefore $P_{i,j}^{(n)} \rightarrow 0$. □

Theorem 5.4 (Basic limit theorem) For an irreducible, recurrent, aperiodic Markov chain, $\lim_{n \rightarrow \infty} P_{i,j}^{(n)}$ exists and is independent of the initial state i . Furthermore,

$$\pi_j := \lim_{n \rightarrow \infty} P_{i,j}^{(n)} = \frac{1}{M_{jj}} = \frac{1}{\mathbb{E}[R_j | X_0 = j]}.$$

If the chain is positive recurrent, then $\{\pi_j\}_{j \geq 0}$ is the unique stationary distribution.

Remark 5.5 For any $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots)^\top$ for which all the limits exist, $\boldsymbol{\pi}$ always satisfies $\boldsymbol{\pi} = \mathbf{P}^\top \boldsymbol{\pi}$. If it also satisfies $\mathbf{1}_{|S|}^\top \boldsymbol{\pi} = 1$, then $\boldsymbol{\pi}$ is a stationary distribution. In other words, any limiting distribution is automatically a stationary distribution. Indeed, by the Chapman-Kolmogorov equation,

$$P_{ij}^{(n)} = \sum_{k \in S} P_{ik}^{(n-1)} P_{kj}.$$

Taking limits on both sides gives

$$\begin{aligned}
\lim_{n \rightarrow \infty} P_{ij}^{(n)} &= \sum_{k=0}^{\infty} \lim_{n \rightarrow \infty} P_{ik}^{(n-1)} P_{kj} \\
&= \sum_{k=0}^{\infty} \pi_k P_{kj}.
\end{aligned}$$

Thus $\boldsymbol{\pi} = \mathbf{P}^\top \boldsymbol{\pi}$.

Lecture 6 — Wednesday, September 24, 2014

The examples above show that, for an irreducible aperiodic discrete-time Markov chain, $\lim_{n \rightarrow \infty} \mathbf{P}^n$ can exist and have identical rows. In that case the limiting distribution does not depend on the initial state or initial distribution. For aperiodic chains with more than one positive recurrent class, $\lim_{n \rightarrow \infty} \mathbf{P}^n$ can exist but need not have identical rows, so the limiting distribution can depend on the initial state. If j is transient, then $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0$ for every i .

The basic limit theorem says that if a Markov chain is irreducible, recurrent, and aperiodic, then $\lim_{n \rightarrow \infty} P_{ij}^{(n)}$ exists and is independent of i . Furthermore,

$$\pi_j = \lim_{n \rightarrow \infty} P_{ij}^{(n)} = \frac{1}{M_{jj}}.$$

For irreducible, aperiodic, and positive recurrent Markov chains, the limiting distribution $\boldsymbol{\pi}$ also represents the fraction of time the chain spends in different states in the long run, and it is independent of state i . The fraction of time spent in state j up to time $n - 1$ is

$$\gamma_{i,j}^{(n)} := \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{1}_{\{X_k=j\}}.$$

Now,

$$P_{i,j}^{(n)} = \mathbb{E} \left[\frac{1}{n} \sum_{k=0}^{n-1} \mathbb{1}_{\{X_k=j\}} \mid X_0 = i \right] = \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{P}(X_k = j \mid X_0 = i) = \frac{1}{n} \sum_{k=0}^{n-1} P_{ij}^{(k)}.$$

Since $P_{ij}^{(k)} \rightarrow \pi_j$ as $k \rightarrow \infty$, we have $\gamma_{i,j}^{(n)} \rightarrow \pi_j$. On the other hand, by the law of large numbers,

$$\frac{n}{\sum_{k=0}^{n-1} \mathbb{1}_{\{X_k=j\}}} = \frac{n}{\text{time spent in state } j \text{ up to time } n-1} \longrightarrow \mathbb{E}[R_j \mid X_0 = i].$$

This implies that

$$\frac{1}{n} \sum_{k=0}^{n-1} \mathbb{1}_{\{X_k=j\}} \rightarrow \frac{1}{M_{jj}}$$

almost surely. Then

$$P_{i,j}^{(n)} = \mathbb{E} \left[\frac{1}{n} \sum_{k=0}^{n-1} \mathbb{1}_{\{X_k=j\}} \mid X_0 = i \right] \longrightarrow \frac{1}{M_{jj}}.$$

We thus conclude that

$$\pi_j = \frac{1}{M_{jj}}.$$

This implies that π is the long-run fraction of time the Markov chain spends in different states.

Remark 6.1

1. For a periodic Markov chain, $\lim_{n \rightarrow \infty} P_{ij}^{(n)}$ does not exist in general, but

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} P_{ij}^{(k)}$$

always exists, by the law of large numbers. Hence $\lim_{n \rightarrow \infty} \gamma_{i,j}^{(n)}$ always exists.

2. In the periodic case, there is an extension of the basic limit theorem:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} P_{ij}^{(k)} = \frac{1}{M_{jj}}.$$

For example, if

$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

then $d = 2$ and $\lim_{n \rightarrow \infty} \mathbf{P}^{2n} = \mathbf{I}$, so in particular $P_{00}^{(2n)} \rightarrow 1$, while $1/M_{00} = 1/2$.

Theorem 6.2 If state i is null recurrent, then $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0$ for any state j .

Proof. If $i = j$, then $\lim_{n \rightarrow \infty} P_{ii}^{(n)} = 0$. Also, $P_{ii}^{(n)} = 0$ when n is not a multiple of the period d , so the full limit is still 0. If $i \neq j$, there are two cases:

1. If i and j are not in the same class, then since i is recurrent, j is not accessible from i . Hence $P_{ij}^{(n)} = 0$ and $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0$.
2. If i and j are in the same class, then j is also null recurrent. Recall that

$$P_{ij}^{(n)} = \sum_{k=1}^n f_{ij}^{(k)} P_{jj}^{(n-k)}.$$

Therefore

$$\lim_{n \rightarrow \infty} P_{ij}^{(n)} = \sum_{k=1}^{\infty} f_{ij}^{(k)} \lim_{n \rightarrow \infty} P_{jj}^{(n-k)} = \sum_{k=1}^{\infty} f_{ij}^{(k)} \cdot 0 = 0.$$

□

Remark 6.3 The second case above also shows that $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0$ for all null recurrent j .

Theorem 6.4 In a finite-state Markov chain, there are no null recurrent states.

Proof. Suppose, for a contradiction, that i is a null recurrent state. By Remark 6.3, $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0$ for every $j \in S$. Therefore

$$1 = \lim_{n \rightarrow \infty} \sum_{j \in S} P_{ij}^{(n)} = \sum_{j \in S} \lim_{n \rightarrow \infty} P_{ij}^{(n)} = \sum_{j \in S} 0 = 0,$$

which is a contradiction. □

Theorem 6.5 An irreducible Markov chain is positive recurrent if and only if a stationary distribution exists.

Proof. *Forward implication.* First assume that the Markov chain is positive recurrent. For state i , define γ by

$$\gamma_j = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} P_{ij}^{(k)}.$$

For any $M = 1, 2, 3, \dots$, let $\mathbf{1}_{M+1}$ denote the column vector whose first $M+1$ entries are equal to 1 and whose remaining entries are 0. Then

$$\sum_{j=0}^M \gamma_j = \gamma^\top \mathbf{1}_{M+1} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbf{e}_i^\top \mathbf{P}^k \mathbf{1}_{M+1} \leq 1,$$

since $\mathbf{P}^k \mathbf{1}_{M+1} \leq \mathbf{1}_{|S|}$ componentwise. Hence

$$\sum_{j=0}^{\infty} \gamma_j = 1.$$

Moreover, because the chain is positive recurrent, the first return time $R_i = \min\{n \geq 1 : X_n = i\}$ has finite mean. By the basic limit theorem,

$$\gamma_j = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} P_{jj}^{(k)} = \frac{1}{M_{jj}} > 0.$$

Thus

$$0 < \sum_{j=0}^{\infty} \gamma_j \leq 1.$$

Now, consider the Césaro means

$$\gamma^{(n)} = \frac{1}{n} \sum_{k=0}^{n-1} (\mathbf{P}^\top)^k \mathbf{e}_i.$$

By definition, $\gamma^{(n)} \rightarrow \gamma$ componentwise. Then

$$\mathbf{P}^\top \gamma^{(n)} = \frac{1}{n} \sum_{k=0}^{n-1} (\mathbf{P}^\top)^{k+1} \mathbf{e}_i = \frac{1}{n} \sum_{k=1}^n (\mathbf{P}^\top)^k \mathbf{e}_i = \frac{1}{n} \sum_{k=1}^{n-1} (\mathbf{P}^\top)^k \mathbf{e}_i + \frac{1}{n} ((\mathbf{P}^\top)^n \mathbf{e}_i - \mathbf{e}_i) = \gamma^{(n)} + \frac{1}{n} ((\mathbf{P}^\top)^n \mathbf{e}_i - \mathbf{e}_i).$$

Taking limits gives

$$\mathbf{P}^\top \gamma = \gamma + \lim_{n \rightarrow \infty} \frac{1}{n} ((\mathbf{P}^\top)^n \mathbf{e}_i - \mathbf{e}_i) = \gamma,$$

since each component of $(\mathbf{P}^\top)^n \mathbf{e}_i - \mathbf{e}_i$ is bounded between -1 and 1 . Renormalize by defining

$$\pi_j = \frac{\gamma_j}{\sum_{i \in S} \gamma_i}$$

with $\pi_j \geq 0$. Then $\mathbf{P}^\top \boldsymbol{\pi} = \boldsymbol{\pi}$ since the system of equations is homogeneous, and $\mathbf{1}_{|S|}^\top \boldsymbol{\pi} = 1$ by construction. Thus $\boldsymbol{\pi}$ is a stationary distribution.

Lecture 7 — Monday, September 29, 2014

Reverse implication. Suppose that a stationary distribution π exists, but the chain is null recurrent or transient. Null recurrence and transience imply that $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0$ for every $i, j \in S$. However, for a stationary distribution π ,

$$\pi_j = \sum_{i \in S} \pi_i P_{ij}^{(n)} \quad \text{for all } j \text{ and } n = 1, 2, 3, \dots$$

Passing to the limit on both sides gives

$$\pi_j = \lim_{n \rightarrow \infty} \sum_{i \in S} \pi_i P_{ij}^{(n)} = \sum_{i \in S} \pi_i \lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0.$$

Since this holds for every j , we have $\pi = (0, 0, \dots)$, which is not a distribution. This contradiction proves the converse direction. $\square$

Theorem 7.1 Let $\{X_n\}_{n \geq 0}$ be an irreducible, positive recurrent Markov chain. Fix $i \in S$. For $j \in S$, define

$$\nu_j = \mathbb{E} \left[\sum_{k=0}^{R_i-1} \mathbb{1}_{\{X_k=j\}} \mid X_0 = i \right]$$

to be the expected number of visits to j before returning to i , given $X_0 = i$. If $\nu = (\nu_1, \nu_2, \dots)$, then

$$\pi = \frac{\nu}{\sum_{j \in S} \nu_j}$$

is the stationary distribution of $\{X_n\}$.

To see why this formula holds, define $R_i(k)$ to be the time of the k th revisit to state i , so $R_i(1) = R_i$. Let $Y_j(k)$ be the number of visits to state j between the $(k-1)$ st and k th visits to i . Then

$$Y_j(k) = \sum_{n=R_i(k-1)}^{R_i(k)-1} \mathbb{1}_{\{X_n=j\}}.$$

By the Markov property, the random variables $Y_j(k)$ are independent and identically distributed.

By the strong law of large numbers, given $X_0 = i$,

$$\frac{1}{m} \sum_{k=1}^m Y_j(k) \rightarrow \mathbb{E}[Y_j(1) \mid X_0 = i] = \nu_j$$

as $m \rightarrow \infty$. On the other hand,

$$\begin{aligned} \lim_{m \rightarrow \infty} \frac{1}{m} \sum_{k=1}^m Y_j(k) &= \lim_{m \rightarrow \infty} \frac{\sum_{n=0}^{R_i(m)-1} \mathbb{1}_{\{X_n=j\}}}{R_i(m)} \cdot \frac{R_i(m)}{m} \\ &= \text{long-run proportion of time spent in } j \cdot M_{ii}. \end{aligned}$$

Thus ν_j is the long-run proportion of time spent in j multiplied by the positive constant M_{ii} . Since the long-run proportions form the stationary distribution, normalizing ν gives the stationary distribution. Notice that

$$\sum_{j \in S} \nu_j = M_{ii}$$

in this setting, and

$$\frac{\nu_j}{\nu_i} = \frac{\pi_j}{\pi_i}.$$

Here is a nice theorem which we note in passing:

Theorem 7.2 (Markov chain ergodic theorem) For a positive recurrent, irreducible Markov chain with stationary distribution π , consider a **reward function** $f : S \rightarrow \mathbb{R}$. If

$$\sum_{j \in S} \pi_j |f(j)| < \infty,$$

then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} f(X_k) = \pi \cdot \mathbf{f}^\top = \sum_{i \in S} \pi_i f(i) \quad \text{almost surely.}$$

That is, the time average reward converges to the stationary average reward.

Reducible Markov Chains

The canonical decomposition expresses the state space S of a Markov chain as

$$S = T \cup C_1 \cup C_2 \cup \dots,$$

where T consists of all transient states, not necessarily forming a single class, and the C_i are closed disjoint classes of recurrent states.

Definition 7.3 A set of states C is **closed** if $P_{ij} = 0$ for all $i \in C$ and $j \in S \setminus C$.

Thus, if $i \in C$ and $j \in S \setminus C$, then $\mathbb{P}(\text{ever visits } j \mid X_0 = i) = 0$. If we reorder the states, the transition matrix $\mathbf{P}$ can be written in block-diagonal form as

$$\mathbf{P} = \begin{pmatrix} \mathbf{P}_1 & \mathbf{0}_{12} & \mathbf{0}_{13} & \cdots \\ \mathbf{0}_{21} & \mathbf{P}_2 & \mathbf{0}_{23} & \cdots \\ \mathbf{0}_{31} & \mathbf{0}_{32} & \mathbf{P}_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

where $\mathbf{P}_1, \mathbf{P}_2, \dots$ are transition probability matrices for the recurrent classes. If the chain starts in a recurrent class C , then one can restrict attention to the chain on C .

Example 7.4 Consider a Markov chain with

$$\mathbf{P} = \begin{pmatrix} p & 0 & 1-p & 0 \\ 0 & r & 0 & 1-r \\ q & 0 & 1-q & 0 \\ 0 & s & 0 & 1-s \end{pmatrix}$$

This chain is not irreducible. There are two classes, $\{0, 2\}$ and $\{1, 3\}$, both positive recurrent. Rearranging the terms gives

$$\mathbf{P}^* = \begin{pmatrix} p & 1-p & 0 & 0 \\ q & 1-q & 0 & 0 \\ 0 & 0 & r & 1-r \\ 0 & 0 & s & 1-s \end{pmatrix} = \begin{pmatrix} \mathbf{P}_1 & \mathbf{0}_{2 \times 2} \\ \mathbf{0}_{2 \times 2} & \mathbf{P}_2 \end{pmatrix}$$

where $\mathbf{P}_1$ is the transition probability matrix for the first class and $\mathbf{P}_2$ is the transition probability matrix for the second class. Then

$$(\mathbf{P}^*)^n = \begin{pmatrix} \mathbf{P}_1^n & 0 \\ 0 & \mathbf{P}_2^n \end{pmatrix}.$$

For the class $\{0, 2\}$,

$$\mathbf{P}_1 = \begin{pmatrix} p & 1-p \\ q & 1-q \end{pmatrix}.$$

Its stationary distribution (π_0, π_2) satisfies

$$(\pi_0, \pi_2)\mathbf{P}_1 = (\pi_0, \pi_2), \quad \pi_0 + \pi_2 = 1.$$

Hence

$$\pi_0(1-p) = \pi_2q, \quad (\pi_0, \pi_2) = \left(\frac{q}{q+1-p}, \frac{1-p}{q+1-p} \right).$$

Therefore

$$\lim_{n \rightarrow \infty} \mathbf{P}_1^n = \begin{pmatrix} \frac{q}{q+1-p} & \frac{1-p}{q+1-p} \\ \frac{q}{q+1-p} & \frac{1-p}{q+1-p} \end{pmatrix}.$$

For the class $\{1, 3\}$,

$$\mathbf{P}_2 = \begin{pmatrix} r & 1-r \\ s & 1-s \end{pmatrix}.$$

Its stationary distribution (π_1, π_3) satisfies

$$(\pi_1, \pi_3)\mathbf{P}_2 = (\pi_1, \pi_3), \quad \pi_1 + \pi_3 = 1,$$

so

$$\pi_1(1-r) = \pi_3s, \quad (\pi_1, \pi_3) = \left(\frac{s}{s+1-r}, \frac{1-r}{s+1-r} \right).$$

Thus

$$\lim_{n \rightarrow \infty} \mathbf{P}_2^n = \begin{pmatrix} \frac{s}{s+1-r} & \frac{1-r}{s+1-r} \\ \frac{s}{s+1-r} & \frac{1-r}{s+1-r} \end{pmatrix}.$$

Returning to the original state order $(0, 1, 2, 3)$, we obtain

$$\lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} \frac{q}{q+1-p} & 0 & \frac{1-p}{q+1-p} & 0 \\ 0 & \frac{s}{s+1-r} & 0 & \frac{1-r}{s+1-r} \\ \frac{q}{q+1-p} & 0 & \frac{1-p}{q+1-p} & 0 \\ 0 & \frac{s}{s+1-r} & 0 & \frac{1-r}{s+1-r} \end{pmatrix}.$$

The stationary distributions of the full reducible chain are not unique. They are exactly the

convex combinations of the two class-stationary distributions:

$$\pi^{(\lambda)} = \left(\lambda \frac{q}{q+1-p}, (1-\lambda) \frac{s}{s+1-r}, \lambda \frac{1-p}{q+1-p}, (1-\lambda) \frac{1-r}{s+1-r} \right), \quad 0 \leq \lambda \leq 1.$$

Lecture 8 — Wednesday, October 01, 2014

Recall that irreducible Markov chain is positive recurrent if and only if a stationary distribution exists. For an irreducible, positive recurrent Markov chain, the long-run proportion of time spent in each state forms a stationary distribution. Thus, if $\{X_n\}$ is irreducible and positive recurrent, define

$$\nu_j = \mathbb{E} \left[\sum_{k=0}^{R_i-1} \mathbb{1}_{\{X_k=j\}} \mid X_0 = i \right].$$

Then

$$\pi_j = \frac{\nu_j}{\sum_{\ell \in S} \nu_\ell}$$

defines the stationary distribution of $\{X_n\}$.

The long-run behaviour is summarized by

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} f(X_i) = \pi \cdot f^T.$$

For reducible Markov chains, the canonical decomposition is $S = T \cup C_1 \cup C_2 \cup \dots$.

In summary, $\lim_{n \rightarrow \infty} P_{ij}^{(n)}$ exists if j is aperiodic. The limit is the long-run proportion of time spent in state j . For any fixed i , these limits form a distribution if and only if there exists at least one positive recurrent class. The limit is independent of i for every j if and only if there is at most one positive recurrent class. The limit is 0 when j is transient, 0 when j is null recurrent, 0 when j is positive recurrent but i and j are not in the same class, and positive when i and j are positive recurrent and in the same class. If i is transient and j is positive recurrent, the value is related to an absorption probability. Once the limit forms a distribution, it is a stationary distribution. A stationary distribution π exists if and only if there is at least one positive recurrent class, and it is unique if and only if there is a unique positive recurrent class. When a stationary distribution exists, $\pi_j = 0$ if j is null recurrent or transient. In particular, for an irreducible Markov chain, the limiting distribution, when it exists by positive recurrence and aperiodicity, agrees with the

stationary distribution and the long-run proportion of time spent in each state.

Simple Random Walk

Definition 8.1 Let $\{X_n\}_{n \geq 1}$ be independent and identically distributed, with

$$\mathbb{P}(X_i = 1) = p = 1 - q = 1 - \mathbb{P}(X_i = -1).$$

Define the **simple random walk** $\{S_n\}_{n \geq 0}$ by $S_0 = 0$ and $S_n = X_1 + \cdots + X_n$.

The simple random walk is a Markov chain because

$$\begin{aligned} \mathbb{P}(S_{n+1} = s_{n+1} \mid S_n = s_n, \dots, S_0 = 0) \\ &= \mathbb{P}(X_{n+1} = s_{n+1} - s_n \mid S_n = s_n, \dots, S_0 = 0) \\ &= \mathbb{P}(X_{n+1} = s_{n+1} - s_n) \\ &= \mathbb{P}(S_{n+1} = s_{n+1} \mid S_n = s_n). \end{aligned}$$

If $p = 1$ or $q = 1$, the chain is deterministic, all states are transient, and each state forms its own equivalence class. If $p \in (0, 1)$, the chain is irreducible with period $d = 2$.

Proposition 8.2 The simple random walk is transient if $p \neq q$, and null recurrent if $p = q = 1/2$.

Proof. If $p \neq q$, assume without loss of generality that $p > q$. By the strong law of large numbers, $S_n \rightarrow \infty$ almost surely. Then there is a last visit to state 0, so state 0 is transient, and the chain is transient. If $p = 1/2$, we show

$$\sum_{n=0}^{\infty} P_{00}^{(2n)} = \infty$$

to prove recurrence. Since $d = 2$, $P_{00}^{(2n+1)} = 0$ for all $n = 0, 1, \dots$. Also,

$$P_{00}^{(2n)} = \mathbb{P}(n \text{ steps left, } n \text{ steps right}) = \binom{2n}{n} \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n = \binom{2n}{n} \left(\frac{1}{4}\right)^n$$

Using Stirling's formula $n! \sim \sqrt{2\pi n} e^{-n} n^n$,

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} \sim \frac{\sqrt{4\pi n} e^{-2n} (2n)^{2n}}{2\pi n e^{-2n} n^{2n}} = \frac{2\sqrt{\pi n} 2^{2n} n^{2n}}{2\pi n n^{2n}} = \frac{2^{2n}}{\sqrt{\pi n}}$$

Then

$$P_{00}^{(2n)} \sim \left(\frac{2^{2n}}{\sqrt{\pi n}} \right) \left(\frac{1}{4} \right)^n = \frac{1}{\sqrt{\pi n}}.$$

Hence

$$\sum_{n=0}^{\infty} P_{00}^{(2n)} \sim \sum_{n=0}^{\infty} \frac{1}{\sqrt{\pi n}} = \infty.$$

Therefore state 0 is recurrent.

On the other hand, solve the system of equations $\mathbf{P}^\top \boldsymbol{\pi} = \boldsymbol{\pi}$. The i th component satisfies

$$\pi_i = \frac{1}{2}\pi_{i-1} + \frac{1}{2}\pi_{i+1},$$

so $\pi_{i+1} - \pi_i = \pi_i - \pi_{i-1}$. Since this holds for all $i \in \mathbb{Z}$, the sequence $\{\pi_i\}$ is arithmetic, and $\pi_i = \pi_0 + id$, where $d = \pi_1 - \pi_0$. The only possible nonnegative arithmetic sequence on all of $\mathbb{Z}$ is constant, so $\pi_i = \pi_0$. If $\pi_0 = 0$, then all $\pi_i = 0$ and

$$\sum_{i \in S} \pi_i \neq 1.$$

If $\pi_0 > 0$, then

$$\sum_{i \in S} \pi_i = \infty \neq 1.$$

Therefore no stationary distribution exists. By [Theorem 6.5](#), the chain cannot be positive recurrent. Since it is recurrent, it must be null recurrent. $\square$

Another useful calculation for the simple random walk is to find

$$f_{00} = \mathbb{P}(\text{the Markov chain visits 0 again} \mid X_0 = 0)$$

and, more generally,

$$f_{i0} = \mathbb{P}(\text{visit 0} \mid X_0 = i).$$

Proposition 8.3 For simple random walk,

$$f_{00} = 2 \min\{p, 1-p\} \quad \text{and} \quad f_{1,0} = \frac{1-p}{p} \quad \text{and} \quad f_{i,0} = \left(\frac{1-p}{p} \right)^i \quad \text{for } i \geq 2.$$

Proof. Assume $p \geq 1/2$, so the drift is nonnegative. For any $i \leq j$, $f_{ij} = 1$. Indeed, if $p = 1/2$, then the chain is null recurrent and hence recurrent, so $i \leftrightarrow j$ and $f_{ij} = 1$. If $p > 1/2$, the strong law of large numbers gives $S_n \rightarrow \infty$, so starting from i , the chain eventually visits every state to the right of i .

In particular, $f_{i,0} = 1$ for $i \leq -1$. Using the Markov property, we have

$$\begin{aligned} f_{00} &= \mathbb{P}(\text{the Markov chain ever visits 0 again} \mid X_0 = 0) \\ &= \mathbb{P}(\text{visit 0} \mid X_0 = 0, X_1 = 1)\mathbb{P}(X_1 = 1 \mid X_0 = 0) \\ &\quad + \mathbb{P}(\text{visit 0} \mid X_0 = 0, X_1 = -1)\mathbb{P}(X_1 = -1 \mid X_0 = 0) \\ &= p \cdot f_{1,0} + (1-p) \cdot f_{-1,0} \\ &= p \cdot f_{1,0} + 1 - p. \end{aligned}$$

Similarly,

$$f_{1,0} = p \cdot f_{2,0} + (1-p) \cdot f_{0,0} \tag{8.1}$$

and for $i \geq 2$,

$$f_{i,0} = p \cdot f_{i+1,0} + (1-p) \cdot f_{i-1,0}. \tag{8.2}$$

From Equation (8.1), rearranging gives

$$p(f_{1,0} - f_{2,0}) = (1-p)(f_{0,0} - f_{1,0}),$$

and from Equation (8.2),

$$p(f_{i,0} - f_{i+1,0}) = (1-p)(f_{i-1,0} - f_{i,0}).$$

Thus, in general,

$$f_{i,0} - f_{i+1,0} = \frac{1-p}{p}(f_{i-1,0} - f_{i,0}).$$

For the simple random walk, if the chain goes from $i > 0$ to 0, it must hit every intermediate state. Therefore

$$f_{i,0} = f_{i,i-1}f_{i-1,i-2} \cdots f_{1,0}.$$

By construction, $f_{j,j-1} = \alpha$ does not depend on j , so $f_{i,0} = \alpha^i$. The differences of consecutive terms also form a geometric sequence with ratio α . Therefore $\alpha = (1-p)/p$, so

$$f_{1,0} = \frac{1-p}{p} \quad \text{and} \quad f_{i,0} = \left(\frac{1-p}{p}\right)^i \quad \text{for } i \geq 2.$$

Then

$$f_{00} = pf_{1,0} + (1-p) = 2(1-p).$$

By symmetry, $f_{00} = 2p$ if $p < 1/2$. Hence

$$f_{00} = 2 \min\{p, 1-p\}.$$

□

Lecture 9 — Monday, October 06, 2014

For any fixed i , the limit $\lim_{n \rightarrow \infty} P_{ij}^{(n)}$ forms a distribution precisely when there is one positive recurrent class and no null recurrent class.

For the simple random walk, the chain is transient if $p \neq q$ and null recurrent if $p = q$. We also found that

$$f_{00} = 2 \min\{p, 1-p\}.$$

G/M/1 Queues

Consider a discrete-time Markov chain with transition probability matrix

$$\mathbf{P} = \begin{pmatrix} 1 - b_0 & b_0 & 0 & 0 & 0 & \dots \\ 1 - (b_0 + b_1) & b_1 & b_0 & 0 & 0 & \dots \\ 1 - (b_0 + b_1 + b_2) & b_2 & b_1 & b_0 & 0 & \dots \\ 1 - (b_0 + b_1 + b_2 + b_3) & b_3 & b_2 & b_1 & b_0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

where $\mathbf{b} = (b_0, b_1, \dots)$ is a probability distribution.

In the notation $G/M/1$, G indicates a general **interarrival time distribution**, M indicates exponentially distributed **service times**, and 1 indicates one server. We are interested in the queue length at arrival times. Let X_n be the number of customers present upon the n th arrival, not including the arriving customer. The distribution $\mathbf{b}$ describes the number of customers served

between two successive arrivals:

$$b_k = \mathbb{P}(\text{the server has completed } k \text{ services between two arrivals}).$$

Assume $b_0 > 0$ and $b_0 + b_1 < 1$; these conditions imply that the chain is irreducible and aperiodic.

We want to find a condition for this chain to be positive recurrent.

Find a stationary distribution $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots)^\top$ satisfying $\boldsymbol{\pi} = \mathbf{P}^\top \boldsymbol{\pi}$:

$$\pi_0 = \left(1 - \sum_{i=0}^0 b_i\right) \pi_0 + \left(1 - \sum_{i=0}^1 b_i\right) \pi_1 + \dots = \sum_{i=0}^{\infty} \pi_i \left(1 - \sum_{j=0}^{i-1} b_j\right) \quad (9.1)$$

$$\pi_1 = b_0 \pi_0 + b_1 \pi_1 + b_2 \pi_2 + \dots \quad (9.2)$$

$$\pi_2 = b_0 \pi_1 + b_1 \pi_2 + \dots \quad (9.3)$$

In general,

$$\pi_k = \sum_{i=0}^{\infty} b_i \pi_{k-1+i}$$

with

$$\sum_{i=0}^{\infty} \pi_i = 1.$$

Assume the solution has the form $\pi_k = r_0^k (1 - r_0)$ for $k = 0, 1, 2, \dots$ and $r_0 \in (0, 1)$. We want to verify this form in [Equations \(9.1\) to \(9.3\)](#), and then determine r_0 . For $k \geq 1$,

$$\pi_k = \sum_{i=0}^{\infty} b_i r_0^{k-1+i} (1 - r_0) = \sum_{i=0}^{\infty} b_i r_0^{k-1+i} (1 - r_0) \implies r_0 = \sum_{i=0}^{\infty} b_i r_0^i.$$

Let B be the random variable with probability mass function $\mathbb{P}(B = k) = b_k$. Its generating function is

$$B(z) = \mathbb{E}[z^B] = \sum_{i=0}^{\infty} z^i b_i.$$

Thus r_0 is a solution of $z = B(z)$, and we want this equation to have a solution in $(0, 1)$. Because r_0 is a solution of $z = B(z)$, it is an intersection of $y = z$ and $y = B(z)$.

The generating function B has the following properties. First, B is continuous. Second,

$$B(0) = \sum_{i=0}^{\infty} z^i b_i \Big|_{z=0} = b_0 > 0 \quad \text{and} \quad B(1) = \sum_{i=0}^{\infty} b_i = 1,$$

while

$$B'(z) = \frac{d}{dz} \left(\sum_{i=0}^{\infty} z^i b_i \right) = \sum_{i=1}^{\infty} i z^{i-1} b_i > 0$$

for $z \in (0, 1)$ and

$$B'(1) = \sum_{i=1}^{\infty} i b_i = \mathbb{E}[B],$$

and

$$B''(z) = \sum_{i=2}^{\infty} i(i-1) z^{i-2} b_i > 0$$

for $z \in (0, 1)$, so B is convex.

If $B'(1) = \mathbb{E}[B] > 1$, then there is a unique solution in $(0, 1)$. If $B'(1) \leq 1$, then there is no solution r_0 in $(0, 1)$.

We want to verify that the proposed solution satisfies (1):

$$\pi_0 \stackrel{?}{=} \sum_{i=0}^{\infty} \pi_i \left(1 - \sum_{j=0}^{i-1} b_j \right).$$

To this end, we have

$$\begin{aligned} \text{RHS} &= \sum_{i=0}^{\infty} \pi_i - \sum_{i=0}^{\infty} \sum_{j=0}^{i-1} \pi_i b_j \\ &= 1 - \sum_{j=0}^{\infty} \sum_{i=j+1}^{\infty} \pi_i b_j \\ &= 1 - \sum_{j=0}^{\infty} b_j \sum_{i=j+1}^{\infty} r_0^i (1 - r_0) \\ &= 1 - \sum_{j=0}^{\infty} b_j r_0^{j+1} \\ &= 1 - B(r_0) \\ &= 1 - r_0 \\ &= \pi_0 \\ &= \text{LHS}. \end{aligned}$$

Thus we can conclude that with the conditions

$$b_0 > 0, b_0 + b_1 < 1, \mathbb{E}[B] > 1,$$

there exists a stationary distribution. Hence the chain is positive recurrent. In general, $r_0 = B(r_0)$ does not have an explicit solution and must be found numerically, for example by the iteration $r_0^{(n+1)} = B(r_0^{(n)})$ after choosing an initial value. As a special case, suppose that $b_k = \alpha^k(1 - \alpha)$ for $k = 0, 1, 2, \dots$. Then

$$B(z) = \sum_{k=0}^{\infty} \alpha^k(1 - \alpha)z^k = (1 - \alpha) \sum_{k=0}^{\infty} (\alpha z)^k = \frac{1 - \alpha}{1 - \alpha z}$$

for $|\alpha z| < 1$. The equation $z = B(z)$ becomes

$$z = \frac{1 - \alpha}{1 - \alpha z} \implies \alpha z^2 - z + (1 - \alpha) = 0 \implies z_1 = 1, z_2 = \frac{1 - \alpha}{\alpha},$$

When $\alpha > 1/2$, the condition $B'(1) > 1$ gives $z_2 \in (0, 1)$ and hence $r_0 = (1 - \alpha)/\alpha$. If $\mathbb{E}[B] \leq 1$, then $\mathbb{E}[B] < 1$ implies that the chain is transient, while $\mathbb{E}[B] = 1$ implies that the chain is null recurrent.

Lecture 10 — Wednesday, October 08, 2014

For the $G/M/1$ queue, let B be the random variable with probability mass function $\mathbb{P}(B = k) = b_k$, representing the number of services completed between two arrivals. If $\mathbb{E}[B] > 1$, then the chain is positive recurrent. The stationary distribution is

$$\pi_k = r_0^k(1 - r_0), \quad k = 0, 1, 2, \dots,$$

where r_0 is the unique solution of $z = B(z)$ in $(0, 1)$. If $\mathbb{E}[B] = 1$, then the chain is null recurrent, while if $\mathbb{E}[B] < 1$, then the chain is transient.

Absorption

Example 10.1 Consider a Markov chain with transition probability matrix

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ \alpha & \beta & \gamma \\ 0 & 0 & 1 \end{pmatrix}$$

where $\alpha, \beta, \gamma > 0$ and $\alpha + \beta + \gamma = 1$. State 1 is transient, while states 0 and 2 are absorbing. Let

$$T = \min\{n \geq 0 \mid X_n = 0 \text{ or } X_n = 2\},$$

the time until absorption. We want to find the absorption probability to state 0,

$$u = \mathbb{P}(X_T = 0 \mid X_0 = 1),$$

the expected time to absorption, $v = \mathbb{E}[T \mid X_0 = 1]$, and the distribution of T .

A first-step analysis gives

$$\begin{aligned} u &= \mathbb{P}(X_T = 0 \mid X_0 = 1) \\ &= \mathbb{P}(X_T = 0 \mid X_1 = 0, X_0 = 1)\mathbb{P}(X_1 = 0 \mid X_0 = 1) \\ &\quad + \mathbb{P}(X_T = 0 \mid X_1 = 1, X_0 = 1)\mathbb{P}(X_1 = 1 \mid X_0 = 1) \\ &\quad + \mathbb{P}(X_T = 0 \mid X_1 = 2, X_0 = 1)\mathbb{P}(X_1 = 2 \mid X_0 = 1) \\ &= \alpha \cdot 1 + \beta \cdot u + \gamma \cdot 0 \\ &= \alpha + \beta u. \end{aligned}$$

Thus

$$u = \frac{\alpha}{1 - \beta}.$$

For the expected time to absorption,

$$\begin{aligned} v &= \mathbb{E}[T \mid X_0 = 1] \\ &= \sum_{i \geq 0} \mathbb{E}[T \mid X_1 = i, X_0 = 1] \cdot \mathbb{P}(X_1 = i \mid X_0 = 1) \\ &= 1 \cdot \alpha + \mathbb{E}[T \mid X_1 = 1, X_0 = 1] \cdot \beta + 1 \cdot \gamma \\ &= \alpha + \gamma + (v + 1)\beta. \end{aligned}$$

Therefore

$$v = \frac{1}{1 - \beta}.$$

For the distribution of T ,

$$\begin{aligned} \mathbb{P}(T = k \mid X_0 = 1) &= \mathbb{P}(X_k = 0 \text{ or } 2, X_{k-1} = 1, \dots, X_1 = 1 \mid X_0 = 1) \\ &= \mathbb{P}(X_k = 0 \text{ or } 2 \mid X_{k-1} = 1, \dots, X_0 = 1) \\ &\quad \cdot \mathbb{P}(X_{k-1} = 1 \mid X_{k-2} = 1, \dots, X_0 = 1) \cdots \mathbb{P}(X_1 = 1 \mid X_0 = 1) \\ &= (\alpha + \gamma) \cdot \beta^{k-1} \\ &= (1 - \beta) \beta^{k-1}, \end{aligned}$$

so T has a geometric distribution with parameter $1 - \beta$.

In general, consider a finite-state Markov chain with states $0, \dots, M - 1, M, \dots, N$, where $0, \dots, M - 1$ are transient and $M, \dots, N$ are absorbing. By the canonical decomposition,

$$\mathbf{P} = \begin{pmatrix} \mathbf{Q} & \mathbf{R} \\ \mathbf{0}_{(N-M+1) \times M} & \mathbf{I}_{N-M+1} \end{pmatrix}.$$

Let

$$T = \min\{n \geq 0 \mid X_n \in \{M, \dots, N\}\}$$

be the time until absorption. For $i = 0, \dots, M - 1$ and $j = M, \dots, N$, the absorption probability is

$$\begin{aligned} U_{ij} &:= \mathbb{P}(X_T = j \mid X_0 = i) \\ &= \sum_{k=0}^{M-1} \mathbb{P}(X_T = j \mid X_1 = k) \cdot \mathbb{P}(X_1 = k \mid X_0 = i) \\ &= \sum_{k=0}^{M-1} \mathbb{P}(X_T = j \mid X_0 = k) \cdot Q_{ik} + \mathbb{P}(X_T = j \mid X_1 = j) \cdot P_{ij} \\ &= \sum_{k=0}^{M-1} U_{kj} Q_{ik} + R_{ij}. \end{aligned}$$

In matrix notation, let $\mathbf{U} = [U_{ij}]$, an $M \times (N - M + 1)$ matrix. Then

$$\mathbf{U} = \mathbf{R} + \mathbf{Q}\mathbf{U} \quad \text{and} \quad \mathbf{U} = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{R}.$$

To prove that $\mathbf{I} - \mathbf{Q}$ is invertible, let $\mathbf{x}$ satisfy

$$(\mathbf{I} - \mathbf{Q})\mathbf{x} = \mathbf{0}_M.$$

Then $\mathbf{Q}\mathbf{x} = \mathbf{x}$, so for every $n \geq 1$,

$$\mathbf{Q}^n \mathbf{x} = \mathbf{x}.$$

Since $\{0, \dots, M-1\}$ is the transient block, we have $\mathbf{Q}^n \rightarrow \mathbf{0}_{M \times M}$ as $n \rightarrow \infty$. Therefore

$$\mathbf{x} = \lim_{n \rightarrow \infty} \mathbf{Q}^n \mathbf{x} = \mathbf{0}_M.$$

Thus $\ker(\mathbf{I} - \mathbf{Q}) = \{\mathbf{0}_M\}$, so $\mathbf{I} - \mathbf{Q}$ is nonsingular, and hence invertible. For $i = 0, \dots, M-1$, the expected time to absorption is

$$V_i = \mathbb{E}[T \mid X_0 = i] = \sum_{j=0}^{M-1} P_{ij}(1 + \mathbb{E}[T \mid X_0 = j]) + \sum_{j=M}^N 1 \cdot P_{ij} = 1 + \sum_{j=0}^{M-1} P_{ij}V_j.$$

In matrix form, if $\mathbf{v} = (V_0, \dots, V_{M-1})^\top$, then

$$\mathbf{v} = \mathbf{1}_M + \mathbf{Q}\mathbf{v}, \quad \mathbf{v} = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{1}_M.$$

We are also interested in the mean number of visits to a transient state k before absorption:

$$S_{ik} := \mathbb{E}\left[\sum_{n=0}^{T-1} \mathbb{1}_{\{X_n=k\}} \mid X_0 = i\right] = \delta_{ik} + \sum_{j=0}^{M-1} \mathbb{E}\left[\sum_{n=1}^{T-1} \mathbb{1}_{\{X_n=k\}} \mid X_1 = j\right] \cdot P_{ij} = \delta_{ik} + \sum_{j=0}^{M-1} S_{jk} \cdot P_{ij}.$$

In matrix notation,

$$\mathbf{S} = \mathbf{I} + \mathbf{Q}\mathbf{S} \quad \text{and} \quad \mathbf{S} = (\mathbf{I} - \mathbf{Q})^{-1}.$$

Notice that

$$\sum_{k=0}^{M-1} S_{ik} = \mathbb{E}\left[\sum_{k=0}^{M-1} \sum_{n=0}^{T-1} \mathbb{1}_{\{X_n=k\}} \mid X_0 = i\right] = \mathbb{E}\left[\sum_{n=0}^{T-1} \sum_{k=0}^{M-1} \mathbb{1}_{\{X_n=k\}} \mid X_0 = i\right] = \mathbb{E}[T \mid X_0 = i] = V_i.$$

This explains why $\mathbf{v} = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{1}_M = \mathbf{S}\mathbf{1}_M$. For $0 \leq i \leq M-1$,

$$W_{ij} = \mathbb{P}(X_n = j \text{ for some } 1 \leq n \leq T-1 \mid X_0 = i) = \mathbb{P}(\text{ever visits } j \mid X_0 = i).$$

Then let $H_j = \min\{n < T : X_n = j\}$ be the first visit to j before absorption. By the Markov property, conditional on $H_j < T$, the expected number of visits to j from that time onward is S_{jj} .

Conditional on $H_j = \infty$, there are no visits to j . Therefore

$$S_{ij} = W_{ij}S_{jj},$$

and hence

$$W_{ij} = \frac{S_{ij}}{S_{jj}}.$$

We now derive the exact distribution of

$$T = \min\{n \geq 0 \mid X_n \in S_A\}.$$

Assume that we start with a general initial distribution

$$\boldsymbol{\alpha}_0 = (\alpha_0, \alpha_1, \dots, \alpha_{M-1}, \alpha_M, \dots, \alpha_N).$$

Let

$$S_T := \{0, \dots, M-1\}, \quad S_A := \{M, \dots, N\}.$$

Then, for $k \geq 1$,

$$\begin{aligned} \mathbb{P}(T = k) &= \sum_{x_0, \dots, x_{k-1} \in S_T} \sum_{x_k \in S_A} \mathbb{P}(X_0 = x_0, \dots, X_k = x_k) \\ &= \sum_{x_0, \dots, x_{k-1} \in S_T} \sum_{x_k \in S_A} \alpha_{x_0} \left(\prod_{i=1}^{k-1} P_{x_{i-1}, x_i} \right) P_{x_{k-1}, x_k} \\ &= \sum_{x_0, \dots, x_{k-1} \in S_T} \alpha_{x_0} \left(\prod_{i=1}^{k-1} P_{x_{i-1}, x_i} \right) \left(\sum_{x_k \in S_A} P_{x_{k-1}, x_k} \right) \\ &= \boldsymbol{\alpha}_0^\top \mathbf{Q}^{k-1} \mathbf{q}. \end{aligned}$$

Here

$$\mathbf{q} = \begin{pmatrix} \sum_{j=M}^N P_{0,j} \\ \vdots \\ \sum_{j=M}^N P_{M-1,j} \end{pmatrix} = \begin{pmatrix} \mathbb{P}(X_1 \in S_A \mid X_0 = 0) \\ \vdots \\ \mathbb{P}(X_1 \in S_A \mid X_0 = M-1) \end{pmatrix}.$$

Lecture 11 — Wednesday, October 15, 2014

Discrete Phase-Type Distributions

For an absorbing chain with

$$\mathbf{P} = \begin{pmatrix} \mathbf{Q} & \mathbf{R} \\ \mathbf{0}_{(N-M+1) \times M} & \mathbf{I}_{N-M+1} \end{pmatrix}$$

and

$$T = \min\{n \in \mathbb{Z}^+ \mid X_n \in \{M, \dots, N\}\},$$

the absorption probability is

$$U_{ij} = \mathbb{P}(X_T = j \mid X_0 = i),$$

with $\mathbf{U} = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{R}$. The expected absorption time $V_i = \mathbb{E}[T \mid X_0 = i]$ satisfies $\mathbf{v} = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{e}_M$. The mean number of visits to a transient state is

$$S_{ik} = \mathbb{E} \left[\sum_{n=0}^{T-1} \mathbb{1}_{\{X_n=k\}} \mid X_0 = i \right],$$

and $\mathbf{S} = (\mathbf{I} - \mathbf{Q})^{-1}$. The exact distribution of T is

$$\mathbb{P}(T = k) = \boldsymbol{\alpha}_0^\top \mathbf{Q}^{k-1} \mathbf{q},$$

where

$$\mathbf{q} = \left(\sum_{j=M}^N P_{0,j}, \dots, \sum_{j=M}^N P_{M-1,j} \right)^T.$$

For example,

$$\mathbb{P}(T = 0) = \alpha_{0,M} + \dots + \alpha_{0,N} = 1 - (\alpha_{0,0}, \dots, \alpha_{0,M-1}).$$

Definition 11.1 Such a distribution is called a **discrete phase-type distribution**, denoted by $T \sim DPH(\boldsymbol{\alpha}_0, \mathbf{Q})$, where $M = \dim(\mathbf{Q})$ is the number of transient states, $\boldsymbol{\alpha}_0$ is the part of the initial distribution corresponding to transient states, and $\mathbf{Q}$ is the transient-to-

transient submatrix. which is symmetric.

Notice that $\alpha_{0,M} + \cdots + \alpha_{0,N} = 1 - (\alpha_{0,0} + \cdots + \alpha_{0,M-1})$ is determined by α_0 . Similarly

$$q_i = \sum_{j=M}^N P_{ij} = 1 - \sum_{j=0}^{M-1} P_{ij} = 1 - \sum_{j=0}^{M-1} Q_{ij},$$

so $\mathbf{q}$ is determined by $\mathbf{Q}$; in particular, $\mathbf{q} = (\mathbf{I} - \mathbf{Q})\mathbf{e}_M$. Thus

$$\mathbf{P} = \begin{pmatrix} \mathbf{Q} & \mathbf{R} \\ \mathbf{0}_{(N-M+1) \times M} & \mathbf{I}_{N-M+1} \end{pmatrix}$$

can be represented, after combining all absorbing states into one, as

$$\begin{pmatrix} \mathbf{Q} & (\mathbf{I} - \mathbf{Q})\mathbf{e}_M \\ \mathbf{0}_{1 \times M} & 1 \end{pmatrix}.$$

Revisiting [Example 10.1](#),

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ \alpha & \beta & \gamma \\ 0 & 0 & 1 \end{pmatrix}.$$

After rearranging states,

$$\mathbf{P} = \begin{pmatrix} \beta & \alpha & \gamma \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

so $\mathbf{Q} = [\beta]$ and $\mathbf{R} = [\alpha \ \gamma]$.

Combining the two absorbing states into one gives

$$\mathbf{P} = \begin{pmatrix} \beta & \alpha + \gamma \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \beta & 1 - \beta \\ 0 & 1 \end{pmatrix}.$$

Let $\alpha_0 = p$, so $\alpha_{0,0} + \alpha_{0,1} = 1 - p$.

Here $\mathbf{Q} = \beta$, so $\mathbf{q} = 1 - \beta$, and therefore

$$\mathbb{P}(T = k) = p\beta^{k-1}(1 - \beta).$$

Also $\mathbb{P}(T = 0) = 1 - p$. We call the distribution of T a **zero-modified geometric distribution**.

In particular, if $p = \beta$, we obtain the ordinary geometric distribution. Thus, for example, $\text{Geo}(1 - p) = \text{DPH}_1(p, p)$.

Now let $T \sim \text{DPH}_M(\alpha_0, \mathbf{Q})$. For $k = 1, 2, \dots$, its cumulative distribution function is

$$\begin{aligned} \mathbb{P}(T \leq k) &= 1 - \mathbb{P}(T > k) \\ &= 1 - \sum_{j=k+1}^{\infty} \mathbb{P}(T = j) \\ &= 1 - \sum_{j=k+1}^{\infty} \alpha_0^\top \mathbf{Q}^{j-1} \mathbf{q} \\ &= 1 - \alpha_0^\top \mathbf{Q}^k (\mathbf{I} - \mathbf{Q})^{-1} \mathbf{q} \\ &= 1 - \alpha_0^\top \mathbf{Q}^k \mathbf{e}_M. \end{aligned}$$

The tail probability is $\mathbb{P}(T > k) = \alpha_0^\top \mathbf{Q}^k \mathbf{e}_M$. Since T is a nonnegative integer-valued random variable,

$$\mathbb{E}[T] = \sum_{k=0}^{\infty} \mathbb{P}(T > k) = \sum_{k=0}^{\infty} \alpha_0^\top \mathbf{Q}^k \mathbf{e}_M = \alpha_0^\top \left(\sum_{k=0}^{\infty} \mathbf{Q}^k \right) \mathbf{e}_M = \alpha_0^\top (\mathbf{I} - \mathbf{Q})^{-1} \mathbf{e}_M = \alpha_0^\top \mathbf{v}$$

where

$$\mathbf{v} = (\mathbb{E}[T|X_0 = 0], \dots, \mathbb{E}[T|X_0 = M - 1])^T.$$

This agrees with the result obtained earlier using a first-step analysis.

The class of discrete phase-type distributions is closed under convolution: the sum of two independent discrete phase-type random variables is again discrete phase-type.

Let $X \sim \text{DPH}_M(\alpha_0, \mathbf{S})$ and $Y \sim \text{DPH}_N(\beta_0, \mathbf{T})$, with X and Y independent. Define $\mathbf{s}' = (\mathbf{I} - \mathbf{S})\mathbf{e}_M$ and $\mathbf{t}' = (\mathbf{I} - \mathbf{T})\mathbf{e}_N$, and let $Z = X + Y$. To see that Z is also discrete phase-type, consider a Markov chain with transition probability matrix

$$\mathbf{P} = \begin{pmatrix} \mathbf{S} & \mathbf{s}'\beta_0^\top & \mathbf{s}'(1 - \beta_0^\top \mathbf{e}_N) \\ \mathbf{0}_{N \times M} & \mathbf{T} & \mathbf{t}' \\ \mathbf{0}_{1 \times M} & \mathbf{0}_{1 \times N} & 1 \end{pmatrix}$$

This ensures that once X is absorbed, the initial phase distribution for Y is β_0 . Let $a_0 :=$

$1 - \boldsymbol{\alpha}_0^\top \mathbf{e}_M = \mathbb{P}(X = 0)$. The transient initial distribution for Z is

$$\boldsymbol{\delta} = \begin{pmatrix} \boldsymbol{\alpha}_0 \\ a_0 \boldsymbol{\beta}_0 \end{pmatrix} = \begin{pmatrix} \boldsymbol{\alpha}_0 \\ (1 - \boldsymbol{\alpha}_0^\top \mathbf{e}_M) \boldsymbol{\beta}_0 \end{pmatrix},$$

where the second block accounts for the case $X = 0$ at time 0, so Z starts directly in the transient phases of Y . (If $\boldsymbol{\alpha}_0^\top \mathbf{e}_M = 1$, then this second block is $\mathbf{0}_N$.) The result is

$$Z \sim DPH_{M+N}(\boldsymbol{\delta}, \mathbf{Q}).$$

By induction, the sum of a finite number of independent and identically distributed discrete phase-type random variables is again discrete phase-type.

The mixture of two discrete phase-type distributions is also discrete phase-type. Let $X \sim DPH_M(\boldsymbol{\alpha}_0, \mathbf{S})$ and $Y \sim DPH_N(\boldsymbol{\beta}_0, \mathbf{T})$. If

$$Z = \begin{cases} X, & \text{with probability } p, \\ Y, & \text{with probability } 1 - p, \end{cases}$$

and the choice between X and Y is independent of their values, then Z is discrete phase-type. To see this, define

$$\tilde{\mathbf{Q}} = \begin{pmatrix} \mathbf{S} & \mathbf{0}_{M \times N} \\ \mathbf{0}_{N \times M} & \mathbf{T} \end{pmatrix}, \quad \tilde{\boldsymbol{\alpha}}_0 = (p\boldsymbol{\alpha}_0, (1-p)\boldsymbol{\beta}_0),$$

and

$$\tilde{\mathbf{q}} = \begin{pmatrix} (\mathbf{I} - \mathbf{S})\mathbf{e}_M \\ (\mathbf{I} - \mathbf{T})\mathbf{e}_N \end{pmatrix},$$

where $\mathbf{e}_M$ and $\mathbf{e}_N$ are the all-ones column vectors of dimensions M and N , respectively.

For $k \geq 1$,

$$\begin{aligned} \mathbb{P}(Z = k) &= p \mathbb{P}(X = k) + (1 - p) \mathbb{P}(Y = k) \\ &= p \boldsymbol{\alpha}_0^\top \mathbf{S}^{k-1} (\mathbf{I} - \mathbf{S}) \mathbf{e}_M + (1 - p) \boldsymbol{\beta}_0^\top \mathbf{T}^{k-1} (\mathbf{I} - \mathbf{T}) \mathbf{e}_N \\ &= \tilde{\boldsymbol{\alpha}}_0^\top \tilde{\mathbf{Q}}^{k-1} \tilde{\mathbf{q}}. \end{aligned}$$

Also,

$$\begin{aligned} \mathbb{P}(Z = 0) &= p \mathbb{P}(X = 0) + (1 - p) \mathbb{P}(Y = 0) \\ &= p (1 - \boldsymbol{\alpha}_0^\top \mathbf{e}_M) + (1 - p) (1 - \boldsymbol{\beta}_0^\top \mathbf{e}_N) \\ &= 1 - \tilde{\boldsymbol{\alpha}}_0^\top \tilde{\mathbf{e}}_{M+N}, \end{aligned}$$

where $\tilde{\mathbf{e}}_{M+N}$ is the all-ones vector in dimension $M + N$. Therefore $Z \sim DPH_{M+N}(\tilde{\alpha}_0, \tilde{\mathbf{Q}})$.

Lecture 12 — Monday, October 20, 2014

A discrete phase-type distribution is the absorption time distribution for a finite-state Markov chain. If $T \sim DPH_m(\alpha_0, \mathbf{Q})$, where α_0 is the initial distribution on the transient states, m is the number of transient states, and $\mathbf{Q}$ is the transient-to-transient transition probability matrix, then

$$\mathbb{P}(T = k) = \alpha_0^\top \mathbf{Q}^{k-1} \mathbf{q}, \quad \mathbb{P}(T = 0) = 1 - \alpha_0^\top \mathbf{e}_m,$$

where $\mathbf{q} = (\mathbf{I} - \mathbf{Q})\mathbf{e}_m$. Its cumulative distribution function is

$$\mathbb{P}(T \leq k) = 1 - \alpha_0^\top \mathbf{Q}^k \mathbf{e}_m,$$

and

$$\mathbb{E}[T] = \alpha_0^\top (\mathbf{I} - \mathbf{Q})^{-1} \mathbf{e}_m = \alpha_0^\top \mathbf{v}, \quad \mathbf{v} = (\mathbf{I} - \mathbf{Q})^{-1} \mathbf{e}_m.$$

We saw that discrete phase-type distributions are stable under independent sums and finite mixtures. Moreover, Any discrete distribution on the non-negative integers with finite support is discrete phase-type. As a result, we can approximate any discrete distribution on $\mathbb{Z}^+$ arbitrarily closely. If $\mathbf{p} = (p_0, p_1, \dots)$ with

$$\sum_{i=0}^{\infty} p_i = 1.$$

then truncations of the form $(p_0, p_1, \dots, p_n, p^*)$ can be made arbitrarily close by taking n large. In this sense, discrete phase-type distributions are dense in the set of all discrete distributions on $\mathbb{Z}^+$. Discrete phase-type representations are also not unique: the same random variable may admit representations $DPH_m(\alpha_0, \mathbf{Q})$ and $DPH_m(\beta_0, \mathbf{Q}')$.

The absorption results can also be used to solve other problems that do not initially look like absorption problems:

1. To find the expected time until the first visit to a state i , only the behaviour before reaching i matters. Thus the transition probabilities out of i do not matter, and one can make state i absorbing by setting $P_{ii} = 1$ and $P_{ij} = 0$ for $i \neq j$.
2. To find the expected number of visits to state j before visiting state i , make state i absorbing by changing the i th row of $\mathbf{P}$ to be $(0, \dots, 0, 1, 0, \dots)$.

This reduces the problem to finding the expected number of visits to state j before absorption, or the probability of visiting state i before visiting state j .

2. Poisson Processes

Definition and Basic Properties

Definition 12.1 A continuous-time stochastic process $\{N(t), t \geq 0\}$ is a **Poisson process** with parameter $\lambda > 0$ if the following hold:

1. $N(0) = 0$.
2. $N(t)$ has **independent increments**: $N(t_2) - N(t_1)$ and $N(t_4) - N(t_3)$ are independent for $0 \leq t_1 < t_2 \leq t_3 < t_4$.
3. For all $0 \leq s < t$, the random variable $N(t) - N(s)$ follows a Poisson distribution with parameter $\lambda(t - s)$. That is,

$$\mathbb{P}(N(t) - N(s) = n) = e^{-\lambda(t-s)} \frac{(\lambda(t-s))^n}{n!}, \quad n = 0, 1, 2, \dots$$

We can regard the Poisson process $\{N(t), t \geq 0\}$ as the number of random points that fall into the interval $[0, t]$. Under this interpretation, Condition (1) holds naturally, and Condition (2) is equivalent to saying that the numbers of points falling into disjoint intervals are independent.

Proposition 12.2 Condition (3) holds if and only if

$$\mathbb{P}(N(h) = 1) = \lambda h + o(h),$$

where

$$\lim_{h \rightarrow 0} \frac{o(h)}{h} = 0$$

and $\mathbb{P}(N(h) \geq 2) = o(h)$, together with **stationary increments**: $N(t+h) - N(t)$ has the same distribution as $N(h)$.

Proof. *Forward implication.* First suppose that $N(t) - N(s) \sim \text{Poisson}(\lambda(t-s))$. Since $\lambda(t-s)$

is only a function of $t - s$, we have stationary increments. Moreover, as $h \rightarrow 0$,

$$\mathbb{P}(N(h) - N(0) = 1) = e^{-\lambda h}(\lambda h) = \left(1 - \lambda h + \frac{(\lambda h)^2}{2} - \dots\right)(\lambda h) = (1 - o(1))\lambda h = \lambda h + o(h)$$

and

$$\begin{aligned}\mathbb{P}(N(h) - N(0) \geq 2) &= 1 - \mathbb{P}(N(h) - N(0) = 1) - \mathbb{P}(N(h) - N(0) = 0) \\ &= 1 - e^{-\lambda h}\lambda h - e^{-\lambda h} \\ &= e^{-\lambda h} \left(e^{\lambda h} - \lambda h - 1 \right) \\ &= e^{-\lambda h} \left(1 + \lambda h + \frac{(\lambda h)^2}{2} + \frac{(\lambda h)^3}{6} + \dots - \lambda h - 1 \right) \\ &= e^{-\lambda h}(o(h)) \\ &= o(h).\end{aligned}$$

Reverse implication. Define $P_n(t) = \mathbb{P}(N(t) = n)$. For $P_0(t)$, we have:

$$\begin{aligned}P_0(t+h) &= \mathbb{P}(N(t+h) = 0) \\ &= \mathbb{P}(N(t) = 0, N(t+h) - N(t) = 0) \\ &= \mathbb{P}(N(t) = 0) \cdot \mathbb{P}(N(t+h) - N(t) = 0) \quad (\text{by independent increments}) \\ &= P_0(t) \cdot (1 - (\lambda h + o(h))) \\ &= P_0(t) - \lambda h P_0(t) + o(h).\end{aligned}$$

Thus

$$P'_0(t) = \lim_{h \rightarrow 0} \frac{P_0(t+h) - P_0(t)}{h} = -\lambda P_0(t).$$

This is a first-order linear ordinary differential equation. Solving for $P_0(t)$ gives the general solution $ce^{-\lambda t}$. Using the boundary condition

$$P_0(0) = \mathbb{P}(N(0) = 0) = 1,$$

we get $P_0(t) = \mathbb{P}(N(t) = 0) = e^{-\lambda t}$.

In general, for $n \geq 1$:

$$\begin{aligned}P_n(t+h) &= \mathbb{P}(N(t+h) = n) \\ &= \mathbb{P}(N(t) = n, N(t+h) - N(t) = 0) \\ &\quad + \mathbb{P}(N(t) = n-1, N(t+h) - N(t) = 1)\end{aligned}$$

$$\begin{aligned}
& + \mathbb{P}(N(t+h) = n, N(t+h) - N(t) \geq 2) \\
& = P_n(t) \cdot P_0(h) + P_{n-1}(t) \cdot P_1(h) + o(h) \\
& = P_n(t) \cdot (1 - \lambda h) + P_{n-1}(t) \cdot \lambda h + o(h).
\end{aligned}$$

Therefore

$$P'_n(t) = -\lambda P_n(t) + \lambda P_{n-1}(t),$$

or equivalently

$$\frac{d}{dt} \left(e^{\lambda t} P_n(t) \right) = \lambda e^{\lambda t} P_{n-1}(t).$$

Integrating from 0 to t gives

$$e^{\lambda t} P_n(t) - P_n(0) = \lambda \int_0^t e^{\lambda u} P_{n-1}(u) \, du.$$

For $n \geq 1$, $P_n(0) = \mathbb{P}(N(0) = n) = 0$, so

$$e^{\lambda t} P_n(t) = \lambda \int_0^t e^{\lambda u} P_{n-1}(u) \, du.$$

For $n = 1$,

$$e^{\lambda t} P_1(t) = \lambda \int_0^t e^{\lambda u} P_0(u) \, du = \lambda \int_0^t 1 \, du = \lambda t,$$

so $P_1(t) = e^{-\lambda t}(\lambda t)$. For an induction step, assume for some $n \geq 2$ that

$$P_{n-1}(t) = e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}.$$

Then

$$e^{\lambda t} P_n(t) = \lambda \int_0^t e^{\lambda u} P_{n-1}(u) \, du = \lambda \int_0^t \frac{(\lambda u)^{n-1}}{(n-1)!} \, du = \frac{(\lambda t)^n}{n!}.$$

Hence

$$P_n(t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad n = 0, 1, 2, \dots$$

Therefore $N(t) \sim \text{Poisson}(\lambda t)$ for every $t \geq 0$. By stationary increments,

$$N(t) - N(s) \stackrel{d}{=} N(t-s), \quad 0 \leq s < t,$$

so

$$N(t) - N(s) \sim \text{Poisson}(\lambda(t-s)).$$

This is exactly Condition (3). □

Lecture 13 — Wednesday, October 22, 2014

Our earlier absorption results can be used to find the expected time until a first visit to state i and the expected number of visits to state j before visiting state i . We now characterize a Poisson process using the interarrival times $X_1, X_2, \dots$, namely the distances between consecutive points. These interarrival times are independent and identically distributed exponential random variables.

First,

$$\mathbb{P}(X_1 > t) = \mathbb{P}(N(t) = 0) = e^{-\lambda t} \implies \mathbb{P}(X_1 \leq t) = 1 - e^{-\lambda t},$$

so $X_1 \sim \text{Exp}(\lambda)$. Also,

$$\mathbb{P}(X_2 > t \mid X_1 = s) = \mathbb{P}(0 \text{ points in } (s, s+t] \mid X_1 = s) = \mathbb{P}(0 \text{ points in } (s, s+t]) = e^{-\lambda t},$$

which implies that $X_2 \sim \text{Exp}(\lambda)$ and moreover, X_1 and X_2 are independent. Repeating the argument for $X_3, X_4, \dots$ shows that $X_1, X_2, \dots$ are independent and identically distributed $\text{Exp}(\lambda)$ random variables. Thus the Poisson process is the counting process of events with independent exponential interarrival times, and $N(t)$ is the total number of events occurring between time 0 and time t .

Combining and Thinning Poisson Processes

Theorem 13.1 Let $\{N_1(t) : t \geq 0\}$ and $\{N_2(t) : t \geq 0\}$ be two Poisson processes with intensities λ_1 and λ_2 respectively. Then $N(t) = N_1(t) + N_2(t)$ is also a Poisson process with intensity $\lambda_1 + \lambda_2$.

Proof. We check the defining properties. First, $N(0) = N_1(0) + N_2(0) = 0$, so Condition (1) is satisfied.

For

$$0 \leq t_1 < t_2 \leq t_3 < t_4,$$

the increment $N(t_2) - N(t_1)$ is the sum of the corresponding increments of N_1 and N_2 , and $N(t_4) - N(t_3)$ is the sum of the corresponding increments over the disjoint interval $(t_3, t_4]$. Since N_1 and N_2 are independent Poisson processes with independent increments, these two increments

of N are independent, which is Condition (2).

Finally, the sum of independent Poisson random variables with parameters $\lambda_1(t-s)$ and $\lambda_2(t-s)$ is Poisson with parameter $(\lambda_1 + \lambda_2)(t-s)$, which is Condition (3). Hence N is a Poisson process with intensity $\lambda_1 + \lambda_2$. $\square$

Theorem 13.2 (Combining) If X and Y are independent Poisson random variables with parameters a and b , respectively, then $X + Y$ is Poisson with parameter $a + b$. In general, if $N_1(t), N_2(t), \dots, N_k(t)$ are independent Poisson processes with intensities $\lambda_1, \dots, \lambda_k$, then

$$N(t) = \sum_{i=1}^k N_i(t)$$

is Poisson with intensity

$$\lambda = \sum_{i=1}^k \lambda_i.$$

Proof. Let $X \sim \text{Poisson}(a)$ and $Y \sim \text{Poisson}(b)$ be independent. The moment generating function of a Poisson random variable with parameter θ is

$$M_{\text{Poisson}(\theta)}(t) = \exp(\theta(e^t - 1)).$$

Hence

$$M_X(t) = \exp(a(e^t - 1)), \quad M_Y(t) = \exp(b(e^t - 1)).$$

By independence,

$$\begin{aligned} M_{X+Y}(t) &= \mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX} e^{tY}] \\ &= \mathbb{E}[e^{tX}] \mathbb{E}[e^{tY}] = M_X(t) M_Y(t) \\ &= \exp((a+b)(e^t - 1)). \end{aligned}$$

This is the moment generating function of $\text{Poisson}(a+b)$, so $X + Y \sim \text{Poisson}(a+b)$.

Now let $N_1, \dots, N_k$ be independent Poisson processes with intensities $\lambda_1, \dots, \lambda_k$, and define

$$N(t) = \sum_{i=1}^k N_i(t).$$

For each fixed $t \geq 0$, the random variables $N_i(t)$ are independent with

$$N_i(t) \sim \text{Poisson}(\lambda_i t).$$

Applying the result above repeatedly yields

$$N(t) \sim \text{Poisson}\left(t \sum_{i=1}^k \lambda_i\right).$$

Also, $N(0) = 0$, and N has independent stationary increments because each N_i has independent stationary increments and the processes are independent across i . Therefore N is a Poisson process with intensity

$$\lambda = \sum_{i=1}^k \lambda_i.$$

□

Theorem 13.3 (Thinning/Splitting) Suppose that each event of a Poisson process $N(t)$ with intensity λ is independently assigned to type 1 with probability p and to type 2 with probability $1 - p$. Let $N_1(t)$ and $N_2(t)$ count the type 1 and type 2 events, respectively. Then N_1 and N_2 are independent Poisson processes with intensities $p\lambda$ and $(1 - p)\lambda$.

Proof. First, $N_1(0) = 0$. The process $N_1(t)$ has independent increments as a direct consequence of the independent increments property of $N(t)$, and it has stationary increments because $N(t)$ has stationary increments and the splitting rule does not change over time. Moreover,

$$\begin{aligned} \mathbb{P}(N_1(h) = 1) &= \mathbb{P}(N_1(h) = 1 \mid N(h) = 1) \cdot \mathbb{P}(N(h) = 1) \\ &\quad + \mathbb{P}(N_1(h) = 1 \mid N(h) \geq 2) \cdot \mathbb{P}(N(h) \geq 2) \\ &= p(\lambda h + o(h)) + \mathbb{P}(N_1(h) = 1 \mid N(h) \geq 2) \cdot o(h) \\ &= p\lambda h + o(h). \end{aligned}$$

Also

$$\mathbb{P}(N_1(h) \geq 2) \leq \mathbb{P}(N(h) \geq 2) = o(h).$$

Thus $N_1(t)$ is a Poisson process with intensity $p\lambda$. Similarly, $N_2(t)$ is a Poisson process with intensity $(1 - p)\lambda$.

Lecture 14 — Monday, October 27, 2014

It remains to show that N_1 and N_2 are independent. Observe that

$$\begin{aligned}
 \mathbb{P}(N_1(t) = m, N_2(t) = n) &= \mathbb{P}(N_1(t) = m, N_2(t) = n \mid N(t) = m + n) \cdot \mathbb{P}(N(t) = m + n) \\
 &= \binom{m+n}{m} p^m (1-p)^n \cdot e^{-\lambda t} \frac{(\lambda t)^{m+n}}{(m+n)!} \\
 &= \frac{(m+n)!}{m!n!} p^m (1-p)^n e^{-\lambda t} \frac{\lambda^{m+n} t^{m+n}}{(m+n)!} \\
 &= \left(\frac{(p\lambda t)^m}{m!} e^{-p\lambda t} \right) \cdot \left(\frac{((1-p)\lambda t)^n}{n!} e^{-(1-p)\lambda t} \right) \\
 &= \mathbb{P}(N_1(t) = m) \cdot \mathbb{P}(N_2(t) = n).
 \end{aligned}$$

Hence $N_1(t)$ and $N_2(t)$ are independent for each fixed t . To obtain independence of the processes, note that for any disjoint intervals, the vectors of increments of N_1 and N_2 are formed by independently marking the points of the corresponding increments of N . Therefore the finite-dimensional vectors

$$(N_1(t_1), \dots, N_1(t_k)) \quad \text{and} \quad (N_2(t_1), \dots, N_2(t_k))$$

are independent for every $0 \leq t_1 < \dots < t_k$. Thus N_1 and N_2 are independent Poisson processes. $\square$

Order Statistics and Simulation

Let $X_1, \dots, X_n$ be independent and identically distributed random variables. The **order statistics** of $X_1, \dots, X_n$ are $X_{(1)} = \min\{X_1, \dots, X_n\}$, $X_{(2)} =$ second smallest of $\{X_1, \dots, X_n\}$, $\dots$, $X_{(n)} = \max\{X_1, \dots, X_n\}$.

Theorem 14.1 Let $\{N(t), t \geq 0\}$ be a Poisson process with intensity λ . Conditional on $N(t) = n$, the points of N in $[0, t]$ are distributed as the order statistics from a sample of size n from the uniform distribution $\text{Unif}(0, t)$. That is,

$$(S_1, \dots, S_n \mid N(t) = n) \stackrel{d}{=} (U_{(1)}, \dots, U_{(n)}),$$

where $U_1, \dots, U_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, t)$.

Proof. Let

$$a_1 < b_1 < a_2 < b_2 < \cdots < a_n < b_n \leq t.$$

$$\begin{aligned} \mathbb{P}(S_i \in (a_i, b_i], i = 1, \dots, n \mid N(t) = n) &= \frac{\mathbb{P}(N(a_1) = 0, N(b_1) - N(a_1) = 1, \dots, N(t) - N(b_n) = 0)}{\mathbb{P}(N(t) = n)} \\ &= \frac{e^{-\lambda t} \lambda^n \prod_{i=1}^n (b_i - a_i)}{e^{-\lambda t} (\lambda t)^n / n!} \\ &= \frac{n!}{t^n} \prod_{i=1}^n (b_i - a_i). \end{aligned}$$

Dividing both sides by $\prod_{i=1}^n (b_i - a_i)$ and taking the limit as $b_i \rightarrow a_i$, we get

$$f_{S_1, \dots, S_n \mid N(t)=n}(a_1, \dots, a_n) = \frac{n!}{t^n} \mathbb{1}_{\{0 < a_1 < \cdots < a_n < t\}},$$

which is the density of $(U_{(1)}, \dots, U_{(n)})$. □

There are two standard ways to simulate the points of a Poisson process on $[0, t]$. One can simulate independent $\text{Exp}(\lambda)$ interarrival times until their sum is greater than t . Alternatively, one can simulate $N \sim \text{Poisson}(\lambda t)$, then simulate N independent $\text{Unif}[0, t]$ points and arrange them in increasing order.

In many cases, it is helpful to define a Poisson process on all of $\mathbb{R}$ rather than on $[0, \infty)$. Then it is no longer natural to count points starting from 0, since 0 is arbitrary. Instead, we view N as a random set function: for $A \subset \mathbb{R}$, let $N(A)$ be the number of points of the Poisson process in A . Most commonly, A is an interval, so $N((a, b])$ is the number of points in $(a, b]$. From a measure-theoretic perspective, N is often called a **Poisson random measure**. If $A \cap B = \emptyset$, then $N(A)$ and $N(B)$ are independent, and $N(A) \sim \text{Poisson}(\lambda|A|)$, where $|A|$ denotes the Lebesgue measure of A .

3. Continuous-Time Markov Chains

Construction and the Discrete Skeleton

Definition 14.2 A stochastic process $\{X(t)\}_{t \geq 0}$ is a **continuous-time Markov chain** if:

1. For $t \geq 0$, $X(t)$ takes possible values in an at most countable set $\{0, 1, 2, \dots\}$.
2. The Markov property holds: for $s, t \geq 0$ and $i, j \in S$,

$$\mathbb{P}(X(t+s) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s) = \mathbb{P}(X(t+s) = j \mid X(s) = i).$$

The probabilistic properties of the future depend only on the current state, not the past. The chain is called **time-homogeneous** if $\mathbb{P}(X(t+s) = j \mid X(s) = i)$ does not depend on s , which is our default setting.

Lecture 15 — Wednesday, October 29, 2014

The time spent in a given state is called the **sojourn time**. To construct a continuous-time Markov chain, we need to answer two questions:

Question 15.1

1. How long does the chain stay in state i ?
2. When the chain leaves state i , which state does it enter next?

To answer Question (1), let T_i be the random time that the chain spends in state i . Then

$$\begin{aligned} \mathbb{P}(T_i > t + s \mid T_i > s) &= \mathbb{P}(X(u) = i \text{ for all } s \leq u \leq s + t \mid X(u) = i \text{ for all } 0 \leq u \leq s) \\ &= \mathbb{P}(X(u) = i \text{ for all } s \leq u \leq s + t \mid X(s) = i) \\ &= \mathbb{P}(X(u) = i \text{ for all } 0 \leq u \leq t \mid X(0) = i) \\ &= \mathbb{P}(T_i > t). \end{aligned}$$

This is the memoryless property, so T_i follows an exponential distribution. Let its parameter be ν_i , so $T_i \sim \text{Exp}(\nu_i)$. Each time the process enters state i , the amount of time it spends there before going to another state is exponentially distributed with mean $1/\nu_i$. If $\nu_i = 0$, then $\mathbb{P}(T_i > t) = 1$

for every t , so $T_i = \infty$ and state i is absorbing.

For Question (2), once the chain leaves state i , the transition probability is independent of the sojourn time T_i . By the Markov property, the transition probability depends only on the current state, and the current state contains no information about the sojourn time. Thus we denote the probability that the chain goes to j when it leaves i by P_{ij} .

A continuous-time Markov chain stays in state i for an exponential random time T_i , then jumps to another state j with probability P_{ij} , then stays in state j for an exponential random time T_j , and so on.

As in the discrete-time case, we have a transition probability matrix $\{P_{ij}\}_{i,j \in S}$. However, $P_{ii} = 0$ because P_{ij} records the next state after leaving i , so

$$\sum_{j \neq i} P_{ij} = 1$$

for all $i \in S$. The discrete-time Markov chain with transition matrix $\{P_{ij}\}_{i,j \in S}$ and the same initial distribution governs all state changes of the continuous-time chain, but does not record sojourn times. This chain is called the **discrete skeleton** (or the **embedded discrete-time Markov chain**) of the continuous-time Markov chain.

Kolmogorov Forward and Backward Equations

We want to understand how

$$P_{ij}(t) = \mathbb{P}(X(t) = j \mid X(0) = i) = \mathbb{P}(X(t+s) = j \mid X(s) = i)$$

changes as a function of t .

For $i \neq j$, define $q_{ij} = \nu_i P_{ij}$. Here ν_i is the rate of leaving i , while P_{ij} is the probability of jumping to j after leaving i . Thus q_{ij} is the “probability flow” from i to j . Notice that

$$\nu_i = \nu_i \sum_{j \neq i} P_{ij} = \sum_{j \neq i} q_{ij}.$$

For small h ,

$$P_{ii}(h) = \mathbb{P}(X(h) = i \mid X(0) = i) = \mathbb{P}(T_i > h) + o(h)$$

because h is small enough that the probability of two or more transitions is $o(h)$. Hence

$$P_{ii}(h) = e^{-\nu_i h} + o(h) = (1 - \nu_i h + o(h)) + o(h) = 1 - \nu_i h + o(h).$$

$$\lim_{h \rightarrow 0} \frac{P_{ii}(h) - 1}{h} = -\nu_i. \quad (15.1)$$

For $i \neq j$ and h ever so small,

$$\begin{aligned} P_{ij}(h) &= \mathbb{P}(X(h) = j \mid X(0) = i) \\ &= P_{ij} \mathbb{P}(T_i \leq h) + o(h) \\ &= P_{ij}(1 - e^{-\nu_i h}) + o(h) \\ &= P_{ij}(\nu_i h + o(h)) + o(h) \\ &= P_{ij} \nu_i h + o(h) \\ &= q_{ij} h + o(h), \end{aligned}$$

so

$$\lim_{h \rightarrow 0} \frac{P_{ij}(h)}{h} = q_{ij}. \quad (15.2)$$

Let $\mathbf{P}(t) = \{P_{ij}(t)\}_{i,j \in S}$ be the transition probability matrix at time t , with $\mathbf{P}(0) = \mathbf{I}$. Combining Equations (15.1) and (15.2) gives

$$\lim_{h \rightarrow 0} \frac{\mathbf{P}(h) - \mathbf{P}(0)}{h} = \begin{pmatrix} -\nu_0 & q_{01} & q_{02} & \cdots \\ q_{10} & -\nu_1 & q_{12} & \cdots \\ q_{20} & q_{21} & -\nu_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} =: \mathbf{R}.$$

Definition 15.2 The matrix $\mathbf{R}$ is called the **generator matrix** of the continuous-time Markov chain.

The generator matrix combines the information about sojourn times and transition probabilities. Knowing $\mathbf{R}$ determines $\boldsymbol{\nu}$ and $\mathbf{P}$, and conversely. If $\mathbf{R} = \{r_{ij}\}_{i,j \in S}$, then

$$r_{ij} = \begin{cases} -\nu_i & i = j \\ q_{ij} & i \neq j \end{cases}.$$

The row sums of $\mathbf{R}$ are always 0.

The Chapman-Kolmogorov equations still hold in continuous time:

$$\begin{aligned} P_{ij}(t+s) &= \mathbb{P}(X(t+s) = j \mid X(0) = i) \\ &= \sum_{k \in S} \mathbb{P}(X(t+s) = j \mid X(s) = k, X(0) = i) \cdot \mathbb{P}(X(s) = k \mid X(0) = i) \\ &= \sum_{k \in S} P_{kj}(t) \cdot P_{ik}(s). \end{aligned}$$

In matrix notation, $\mathbf{P}(t+s) = \mathbf{P}(s)\mathbf{P}(t)$.

Thus

$$\mathbf{P}(t+h) = \mathbf{P}(h)\mathbf{P}(t),$$

so

$$\mathbf{P}(t+h) - \mathbf{P}(t) = (\mathbf{P}(h) - \mathbf{I})\mathbf{P}(t).$$

Dividing by h and taking $h \rightarrow 0$ gives

$$\mathbf{P}'(t) = \mathbf{R}\mathbf{P}(t).$$

This is the **Kolmogorov backward equation**.

Lecture 16 — Monday, November 03, 2014

Recall that the sojourn time in state i has distribution $T_i \sim \text{Exp}(\nu_i)$. Conditional on leaving state i , the probability of going to j is P_{ij} . The matrix $\{P_{ij}\}$ is a transition probability matrix with $P_{ii} = 0$ for all $i \in S$. The discrete skeleton, also called the embedded discrete-time Markov chain, has transition probability matrix $\{P_{ij}\}$ and the same initial distribution as the continuous-time chain. Also, $\mathbf{P}(t) = \{P_{ij}(t)\}_{i,j \in S}$, and

$$\mathbf{R} = \begin{pmatrix} -\nu_0 & q_{01} & \cdots \\ q_{10} & -\nu_1 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

is the generator matrix, where $q_{ij} = \nu_i P_{ij}$ and

$$\sum_{j \neq i} R_{ij} = 0 \quad \text{for all } i \in S.$$

The Chapman-Kolmogorov equation is $\mathbf{P}(t+s) = \mathbf{P}(t)\mathbf{P}(s)$, and from it we derived the Kolmogorov backward equation $\mathbf{P}'(t) = \mathbf{R}\mathbf{P}(t)$. Similarly,

$$\mathbf{P}(t+h) = \mathbf{P}(t)\mathbf{P}(h) \implies \lim_{h \rightarrow 0} \frac{\mathbf{P}(t+h) - \mathbf{P}(t)}{h} = \mathbf{P}(t) \lim_{h \rightarrow 0} \frac{\mathbf{P}(h) - \mathbf{I}}{h} \implies \mathbf{P}'(t) = \mathbf{P}(t)\mathbf{R}.$$

This is the **Kolmogorov forward equation**.

In entrywise form, the backward equation is

$$P'_{ij}(t) = \sum_{k \in S} R_{ik} P_{kj}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - \nu_i P_{ij}(t).$$

and the forward equation is

$$P'_{ij}(t) = \sum_{k \in S} P_{ik}(t) q_{kj} - \nu_j P_{ij}(t).$$

Remark 16.1 In the derivation of the Kolmogorov forward equation, we quietly interchanged the order of a limit and a summation:

$$\lim_{h \rightarrow 0} \frac{1}{h} (\mathbf{P}(h) - \mathbf{I}) \mathbf{P}(t) = \left(\lim_{h \rightarrow 0} \frac{\mathbf{P}(h) - \mathbf{I}}{h} \right) \mathbf{P}(t).$$

This can be justified if the state space S is finite, or when working with the backward equation. The details involve the dominated convergence theorem from measure theory. In this sense, the backward equation is more fundamental, but the forward equation is often easier to solve.

The matrix $\mathbf{P}(t)$ solves the differential equation $\mathbf{P}'(t) = \mathbf{R}\mathbf{P}(t)$ with initial condition $\mathbf{P}(0) = \mathbf{I}$. The solution is the **matrix exponential**

$$e^{t\mathbf{R}} := \mathbf{I} + t\mathbf{R} + \frac{t^2\mathbf{R}^2}{2!} + \frac{t^3\mathbf{R}^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{R}^n.$$

The series converges, for example by using an operator norm. Moreover, $e^{t\mathbf{R}}$ solves a matrix differential equation:

$$\frac{d}{dt} e^{t\mathbf{R}} = \frac{d}{dt} \left(\mathbf{I} + \sum_{n=1}^{\infty} \frac{t^n}{n!} \mathbf{R}^n \right) = \sum_{n=1}^{\infty} \frac{t^{n-1}}{(n-1)!} \mathbf{R}^n = \mathbf{R} \sum_{n=1}^{\infty} \frac{t^{n-1}}{(n-1)!} \mathbf{R}^{n-1} = \mathbf{R} \sum_{m=0}^{\infty} \frac{t^m}{m!} \mathbf{R}^m = \mathbf{R} e^{t\mathbf{R}}.$$

Also, $e^{t\mathbf{R}}|_{t=0} = \mathbf{I}$. Hence $\mathbf{P}(t) = e^{t\mathbf{R}}$ satisfies the backward equation. The forward equation is handled similarly.

Warning The matrix exponential $e^{t\mathbf{R}}$ is *not* obtained by exponentiating each entry of $t\mathbf{R}$ separately!

In general, $e^{t\mathbf{R}}$ is not easy to calculate. However, if $\mathbf{R}$ is diagonalizable, say $\mathbf{R} = \mathbf{B}\mathbf{D}\mathbf{B}^{-1}$ with $\mathbf{B}$ invertible and $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$, then we have

$$e^{t\mathbf{R}} = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{R}^n = \sum_{n=0}^{\infty} \frac{t^n}{n!} (\mathbf{B}\mathbf{D}\mathbf{B}^{-1})^n = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{B}\mathbf{D}^n\mathbf{B}^{-1} = \mathbf{B} \left(\sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{D}^n \right) \mathbf{B}^{-1} = \mathbf{B} e^{t\mathbf{D}} \mathbf{B}^{-1}$$

where $e^{t\mathbf{D}}$ is computed entrywise along the diagonal:

$$e^{t\mathbf{D}} = \text{diag}(e^{t\lambda_1}, \dots, e^{t\lambda_n}),$$

which holds because for each $m \geq 0$,

$$\mathbf{D}^m = \text{diag}(\lambda_1^m, \dots, \lambda_n^m),$$

and therefore

$$\begin{aligned} e^{t\mathbf{D}} &= \sum_{m=0}^{\infty} \frac{t^m}{m!} \mathbf{D}^m \\ &= \sum_{m=0}^{\infty} \frac{t^m}{m!} \text{diag}(\lambda_1^m, \dots, \lambda_n^m) \\ &= \text{diag} \left(\sum_{m=0}^{\infty} \frac{(t\lambda_1)^m}{m!}, \dots, \sum_{m=0}^{\infty} \frac{(t\lambda_n)^m}{m!} \right) \\ &= \text{diag}(e^{t\lambda_1}, \dots, e^{t\lambda_n}). \end{aligned}$$

Birth and Death Processes

A **birth and death process** is a continuous-time stochastic process $X(t)$, which we think of as the number of individuals in a system at time t . At each time, the population can only change by 1 or -1 , corresponding to a birth or a death. Thus $P_{ij} = 0$ if $|i - j| > 1$, equivalently $q_{ij} = 0$ if

$|i - j| > 1$. The generator matrix $\mathbf{R}$ has the form

$$\mathbf{R} = \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 & \dots \\ \mu_1 & -(\lambda_1 + \mu_1) & \lambda_1 & \dots \\ 0 & \mu_2 & -(\lambda_2 + \mu_2) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix},$$

where $\{\lambda_i\}_{i \geq 0}$ are the birth rates and $\{\mu_i\}_{i \geq 1}$ are the death rates. For $i \geq 1$,

$$q_{i,i+1} = \nu_i P_{i,i+1} = \lambda_i \quad \text{and} \quad q_{i,i-1} = \nu_i P_{i,i-1} = \mu_i.$$

Thus $\nu_i = \lambda_i + \mu_i$. For $i = 0$, $P_{0,1} = 1$. Intuitively, when the system is in state i , the next potential birth occurs after an exponential time with intensity λ_i , and the next potential death occurs after an exponential time with intensity μ_i . These two times “compete” with each other; the smaller time determines the next state change.

If $T_i^{(b)} \sim \text{Exp}(\lambda_i)$ and $T_i^{(d)} \sim \text{Exp}(\mu_i)$ are independent, then

$$\mathbb{P}(T_i^{(b)} < T_i^{(d)}) = \frac{\lambda_i}{\lambda_i + \mu_i} = P_{i,i+1}$$

and

$$\mathbb{P}(T_i^{(d)} < T_i^{(b)}) = \frac{\mu_i}{\lambda_i + \mu_i} = P_{i,i-1}$$

Also, $T_i = T_i^{(b)} \wedge T_i^{(d)}$ is the time until the system changes state, and $T_i \sim \text{Exp}(\lambda_i + \mu_i)$ for $i \geq 1$. Hence $\nu_i = \lambda_i + \mu_i$ for $i \geq 1$, while $\nu_0 = \lambda_0$.

Stationary Distributions for Birth and Death Processes

For a continuous-time Markov chain, a distribution $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots)^\top$ is stationary if $\mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_{|S|}$ and

$$\sum_{j \in S} \pi_j = 1.$$

Recall that for a discrete-time Markov chain, $\mathbf{P}^\top \boldsymbol{\pi} = \boldsymbol{\pi}$. Here, the condition $\mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_{|S|}$ is equivalent to

$$\sum_{i \in S} \pi_i r_{ij} = 0 \quad \text{for all } j \in S.$$

For a birth and death process, we get that for $j = 0$,

$$-\lambda_0 \pi_0 + \mu_1 \pi_1 = 0 \implies \pi_1 = \frac{\lambda_0}{\mu_1} \pi_0$$

and for $j = 1$,

$$\lambda_0\pi_0 - (\lambda_1 + \mu_1)\pi_1 + \mu_2\pi_2 = 0 \implies \pi_2 = \frac{\lambda_1}{\mu_2}\pi_1 = \frac{\lambda_1\lambda_0}{\mu_2\mu_1}\pi_0.$$

In general, for $j \geq 1$,

$$\pi_j = \pi_0 \prod_{k=0}^{j-1} \frac{\lambda_k}{\mu_{k+1}}$$

The condition

$$\sum_{j \in S} \pi_j = 1.$$

implies

$$\pi_0 \left(1 + \sum_{j=1}^{\infty} \prod_{k=0}^{j-1} \frac{\lambda_k}{\mu_{k+1}} \right) = 1$$

This sum must converge for a stationary distribution to exist.

Lecture 17 — Wednesday, November 05, 2014

Recall the Kolmogorov forward equation $\mathbf{P}'(t) = \mathbf{P}(t)\mathbf{R}$. In entrywise form, the backward equation is

$$P'_{ij}(t) = \sum_{k \neq i} q_{ik}P_{kj}(t) - \nu_i P_{ij}(t).$$

The forward equation is

$$P'_{ij}(t) = \sum_{k \neq j} P_{ik}(t)q_{kj} - \nu_j P_{ij}(t).$$

Also, $\mathbf{P}(t) = e^{t\mathbf{R}}$. If $\mathbf{R}$ is diagonalizable, $\mathbf{R} = \mathbf{B}\mathbf{D}\mathbf{B}^{-1}$ for some diagonal $\mathbf{D}$, then $e^{t\mathbf{R}} = \mathbf{B}e^{t\mathbf{D}}\mathbf{B}^{-1}$, where $e^{t\mathbf{D}} = \text{diag}(e^{t\lambda_1}, \dots, e^{t\lambda_n})$. For a birth and death process,

$$P_{ij} = 0 \text{ for } |i - j| > 1 \quad \text{if and only if} \quad q_{ij} = 0 \text{ for } |i - j| > 1.$$

In this case,

$$\mathbf{R} = \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 \\ \mu_1 & -(\lambda_1 + \mu_1) & \lambda_1 \\ 0 & \mu_2 & \ddots \end{pmatrix}.$$

Here $\nu_i = \lambda_i + \mu_i$ and $\nu_0 = \lambda_0$, with λ_i the birth rates and μ_i the death rates. Moreover,

$$P_{i,i+1} = \frac{\lambda_i}{\lambda_i + \mu_i}, \quad P_{i,i-1} = \frac{\mu_i}{\lambda_i + \mu_i}, \quad P_{0,1} = 1.$$

Because of this structure, it is usually easier to write $\mathbf{R}$ directly than to write ν_i and P_{ij} first.

Special Cases of Birth and Death Processes

We enumerate several special cases.

1. If $\mu_i = 0$ for $i = 0, 1, 2, \dots$, this is called a **pure birth process**.
2. More depressingly, if $\lambda_i = 0$ for $i = 0, 1, 2, \dots$, then there are no births, only deaths. This is called a **pure death process**.
3. If $\mu_i = 0$ and $\lambda_i = \lambda > 0$, then the times between births are independent and exponentially distributed with the same parameter λ , regardless of i . This is the counting process of Poisson events, and hence a Poisson process.
4. If $\mu_i = 0$ and $\lambda_i = i\lambda$, this is a **Yule process**. The birth rate is proportional to the current population. Intuitively, each individual has the same constant birth rate λ , independently of the others. Thus the time until the next birth, given current population i , is the minimum of i independent $\text{Exp}(\lambda)$ random variables, hence has distribution $\text{Exp}(i\lambda)$.
5. If $\lambda_i = i\lambda$ and $\mu_i = i\mu$, this is a **linear growth model**. Individuals are independent and have common birth rate λ and death rate μ .
6. If $\lambda_i = \lambda$ for all $i \geq 0$ and

$$\mu_i = \begin{cases} i\mu, & i < s \\ s\mu, & i \geq s \end{cases},$$

this is an **M/M/s queue**. Each arrival is a birth, and each service completion is a death. For $i < s$, there are i busy servers, so the death rate is the sum of the individual service rates, namely $i\mu$. For $i \geq s$, all s servers are busy, and the death rate is $s\mu$.

7. If $\lambda_i = \lambda$ for all $i \geq 0$ and $\mu_i = \mu$ for all $i \geq 1$, this is the special case of an $M/M/s$ queue with $s = 1$, namely an **M/M/1 queue**.

Consider the forward equations for birth and death processes, $\mathbf{P}'(t) = \mathbf{P}(t)\mathbf{R}$. For the special case of a pure birth process, where $\mu_i = 0$ for $i = 1, 2, \dots$, we have $P_{ij}(t) = 0$ if $j < i$. If $j = i$, then

$P'_{ii}(t) = -\lambda_i P_{ii}(t)$, so $P_{ii}(t) = e^{-\lambda_i t}$. Probabilistically, since the chain is a pure birth process,

$$P_{ii}(t) = \mathbb{P}(X(t) = i \mid X(0) = i) = \mathbb{P}(T_i > t) = e^{-\lambda_i t}.$$

For $j > i$,

$$\begin{aligned} P'_{ij}(t) &= \lambda_{j-1} P_{i,j-1}(t) - \lambda_j P_{ij}(t), \\ P'_{ij}(t) + \lambda_j P_{ij}(t) &= \lambda_{j-1} P_{i,j-1}(t). \end{aligned}$$

Multiplying by $e^{\lambda_j t}$ and integrating from 0 to t gives

$$\begin{aligned} \frac{d}{dt} \left(e^{\lambda_j t} P_{ij}(t) \right) &= \lambda_{j-1} e^{\lambda_j t} P_{i,j-1}(t), \\ e^{\lambda_j t} P_{ij}(t) - P_{ij}(0) &= \lambda_{j-1} \int_0^t e^{\lambda_j s} P_{i,j-1}(s) \, ds. \end{aligned}$$

Since $j > i$, we have $P_{ij}(0) = 0$. Therefore

$$P_{ij}(t) = \lambda_{j-1} e^{-\lambda_j t} \int_0^t e^{\lambda_j s} P_{i,j-1}(s) \, ds,$$

which is the recursive formula for $P_{ij}(t)$ when $i < j$. Now consider the special case

$$\lambda_i = \lambda \quad (i = 0, 1, 2, \dots), \quad \mu_i = 0 \quad (i = 1, 2, \dots).$$

This is the Poisson process case from Case 3. above. We show that

$$P_{i,i+k}(t) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots$$

by induction. For $k = 0$, we have $P_{ii}(t) = e^{-\lambda t}$. Assume that for some $k \geq 0$,

$$P_{i,i+k}(t) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}.$$

Using the recursive formula,

$$\begin{aligned} P_{i,i+k+1}(t) &= \lambda \int_0^t e^{-\lambda(t-s)} P_{i,i+k}(s) \, ds \\ &= \lambda e^{-\lambda t} \int_0^t e^{\lambda s} \frac{e^{-\lambda s} (\lambda s)^k}{k!} \, ds \\ &= \frac{e^{-\lambda t} \lambda^{k+1}}{k!} \int_0^t s^k \, ds \end{aligned}$$

$$\begin{aligned}
&= \frac{e^{-\lambda t} \lambda^{k+1}}{k!} \cdot \frac{t^{k+1}}{k+1} \\
&= e^{-\lambda t} \frac{(\lambda t)^{k+1}}{(k+1)!}.
\end{aligned}$$

Continuous-Time Recurrence and Stationarity

Recall that if the continuous-time chain $\{X(t)\}_{t \geq 0}$ is observed only when the state changes, the resulting process is a discrete-time Markov chain $\{X_n\}_{n=0}^\infty$, called the discrete skeleton or embedded discrete-time Markov chain. Its transition probability matrix is given by

$$P_{ij} = \mathbb{P}(X(t_i) = j \mid X(0) = i).$$

We now ask how the discrete-time notions translate into continuous time.

Lecture 18 — Monday, November 10, 2014

The special cases of birth and death processes include pure death processes, pure birth processes, Poisson processes, Yule processes, linear growth models, and $M/M/s$ queues. For a pure birth process, the forward equation gives the recursive formula

$$P_{ij}(t) = \lambda_{j-1} e^{-\lambda_j t} \int_0^t e^{\lambda_j s} P_{i,j-1}(s) ds.$$

For special cases such as $\lambda_i = \lambda$ and $\mu_i = 0$, a closed-form solution can be found.

Once we have $\mathbf{P}(t) = \{P_{ij}(t)\}$, the distribution of $X(t)$ can be expressed as in the discrete-time case:

$$\mathbb{P}(X(t) = j) = \sum_{i \in S} \mathbb{P}(X(t) = j \mid X(0) = i) \mathbb{P}(X(0) = i) = \sum_{i \in S} \alpha_i P_{ij}(t) = (\alpha \mathbf{P}(t))_j.$$

That is, $\alpha_t = \mathbf{P}(t)^\top \alpha_0$.

The transition probability matrix is determined by $\mathbf{R}$. If $f : S \rightarrow \mathbb{R}$ is a function on the state space, then

$$\mathbb{E}[f(X(t))] = \sum_{j \in S} f(j) \mathbb{P}(X(t) = j) = \alpha_t^\top \mathbf{f} = \alpha_0^\top \mathbf{P}(t) \mathbf{f},$$

where $\mathbf{f} = (f(0), f(1), \dots)^\top$ is viewed as a column vector. The matrix $\mathbf{P}(t)$ is called the **transition kernel**.

Definition 18.1 State j is **accessible** from state i , written $i \rightarrow j$, if

$$P_{ij}(t) = \mathbb{P}(X(t) = j \mid X(0) = i) > 0$$

for some $t \geq 0$. States i and j **communicate**, written $i \leftrightarrow j$, if each is accessible from the other. A continuous-time Markov chain is **irreducible** if all states communicate with one another.

Assume $\nu_i > 0$ for every $i \in S$, so that there are no absorbing states created by zero holding rates. Then i and j communicate in continuous time if and only if they communicate in the discrete skeleton. Hence, under this assumption, the continuous-time Markov chain is irreducible if and only if its discrete skeleton is irreducible.

Definition 18.2 Let R_i be the time until the continuous-time Markov chain revisits state i , given $X(0) = i$, and set $R_i = \infty$ if the chain never revisits i . State i is *recurrent* if $\mathbb{P}(R_i < \infty) = 1$, and **transient** if $\mathbb{P}(R_i = \infty) > 0$. A recurrent state is **positive recurrent** if $\mathbb{E}[R_i] < \infty$ and **null recurrent** if $\mathbb{E}[R_i] = \infty$.

The continuous-time Markov chain revisits state i if and only if its discrete skeleton revisits state i . Therefore recurrence and transience agree for a continuous-time Markov chain and its discrete skeleton. There is no analogous notion of periodicity in continuous time, since exponential sojourn times can be arbitrarily small or arbitrarily large.

Example 18.3 Consider the discrete-time Markov chain with transition matrix

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & \dots \\ q & 0 & p & 0 & \dots \\ q & 0 & 0 & p & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

For this chain,

$$f_{00}^{(1)} = 0 \quad \text{and} \quad f_{00}^{(n)} = p^{n-2}q \quad \text{for } n \geq 2.$$

Hence

$$f_{00} = \sum_{n=2}^{\infty} f_{00}^{(n)} = q \sum_{n=2}^{\infty} p^{n-2} = \frac{q}{1-p} = 1.$$

Moreover, the expected number of transitions required to return to 0 is

$$\sum_{n=2}^{\infty} n f_{00}^{(n)} = \sum_{n=2}^{\infty} n p^{n-2} q = 2 \sum_{n=2}^{\infty} p^{n-2} q + \sum_{n=2}^{\infty} (n-2) p^{n-2} q < \infty.$$

Thus the embedded discrete-time chain is positive recurrent.

Now construct a continuous-time Markov chain whose discrete skeleton is this chain. Let W be the number of transitions required to return to 0. If $T_i \sim \text{Exp}(\nu_i)$ is the sojourn time in state i , then the return time to 0 is

$$R_{00} = T_0 + T_1 + \cdots + T_{W-1}.$$

Therefore

$$\begin{aligned} \mathbb{E}[R_{00}] &= \mathbb{E} \left[\mathbb{E} \left[\sum_{i=0}^{W-1} T_i \mid W \right] \right] \\ &= \sum_{n=2}^{\infty} \mathbb{P}(W = n) \cdot \mathbb{E} \left[\sum_{i=0}^{n-1} T_i \mid W = n \right] \\ &= \sum_{n=2}^{\infty} \mathbb{P}(W = n) \cdot \sum_{i=0}^{n-1} \mathbb{E}[T_i] \\ &= \sum_{i=0}^{\infty} \mathbb{E}[T_i] \cdot \sum_{n=i+1}^{\infty} \mathbb{P}(W = n) \\ &= \sum_{i=0}^{\infty} \mathbb{E}[T_i] \cdot \mathbb{P}(W \geq i+1) \\ &= \frac{1}{\nu_0} + \frac{1}{\nu_1} + \sum_{i=2}^{\infty} \frac{p^{i-1}}{\nu_i}. \end{aligned}$$

By choosing the holding rates $\{\nu_i\}_{i \geq 0}$ so that the sojourn times increase sufficiently quickly far from 0, this expectation can be made infinite; for example, taking $\nu_i = p^i$ for large i makes the final series diverge. Thus a continuous-time Markov chain and its discrete skeleton need not have the same positive recurrence or null recurrence behaviour.

Lecture 19 — Wednesday, November 12, 2014

As in discrete time, recurrence classification is a class property. In an irreducible continuous-time Markov chain, all states are positive recurrent, all states are null recurrent, or all states are transient.

Definition 19.1 Let $\{X(t)\}_{t \geq 0}$ be a continuous-time Markov chain. A column vector $\boldsymbol{\pi} = (\pi_0, \pi_1, \pi_2, \dots)^\top$ with $\pi_i \geq 0$ for all $i \in S$ is a **stationary distribution** if

1. $\mathbf{1}_{|S|}^\top \boldsymbol{\pi} = 1$;
2. $\boldsymbol{\pi} = \mathbf{P}(t)^\top \boldsymbol{\pi}$ for all $t \geq 0$.

Remark 19.2 This distribution is stationary for the same reason as in discrete time. If the initial distribution is $\boldsymbol{\alpha}_0 = \boldsymbol{\pi}$, then

$$\boldsymbol{\alpha}_t = \mathbf{P}(t)^\top \boldsymbol{\alpha}_0 = \mathbf{P}(t)^\top \boldsymbol{\pi} = \boldsymbol{\pi}.$$

Thus the distribution of $X(t)$ is $\boldsymbol{\pi}$ for every $t \geq 0$.

The stationary condition can be written in terms of the generator. If $\boldsymbol{\pi} = \mathbf{P}(t)^\top \boldsymbol{\pi}$ for all $t \geq 0$, then

$$(\mathbf{P}(t)^\top - \mathbf{I})\boldsymbol{\pi} = \mathbf{0}_{|S|}.$$

Dividing by t and taking $t \rightarrow 0$ gives

$$\mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_{|S|}.$$

Conversely, if $\mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_{|S|}$ and $\boldsymbol{\alpha}_0 = \boldsymbol{\pi}$, then, using the backward equation,

$$(\boldsymbol{\alpha}_t)' = ((\mathbf{P}(t)^\top \boldsymbol{\alpha}_0))' = (\mathbf{P}'(t))^\top \boldsymbol{\alpha}_0 = \mathbf{P}(t)^\top \mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_{|S|}.$$

Hence $\boldsymbol{\alpha}_t$ is constant in t , and $\boldsymbol{\pi}$ is stationary. Thus $\boldsymbol{\pi}$ is a stationary distribution if and only if

$$\mathbf{1}_{|S|}^\top \boldsymbol{\pi} = 1 \quad \text{and} \quad \mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_{|S|}.$$

Let $\boldsymbol{\psi}$ be a stationary distribution for the discrete skeleton. Comparing $\mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_{|S|}$ with $\mathbf{P}^\top \boldsymbol{\psi} = \boldsymbol{\psi}$

gives a useful correspondence. The j th component of $\mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_{|S|}$ is

$$\pi_j \nu_j = \sum_{i \neq j} \pi_i q_{ij}.$$

Since $q_{ij} = \nu_i P_{ij}$, this becomes

$$\pi_j \nu_j = \sum_{i \neq j} \pi_i \nu_i P_{ij}.$$

For the discrete skeleton, we have $\psi_j = \sum_{i \in S} \psi_i P_{ij}$, and since $P_{ii} = 0$, we can write

$$\psi_j = \sum_{i \neq j} \psi_i P_{ij}.$$

Notice that $\boldsymbol{\psi}$ satisfies the same system of equations as $\boldsymbol{\pi} \odot \boldsymbol{\nu} = (\pi_0 \nu_0, \pi_1 \nu_1, \dots)$. That is, if $\boldsymbol{\pi}$ is stationary for a continuous-time Markov chain, then $\boldsymbol{\psi} = \boldsymbol{\pi} \odot \boldsymbol{\nu}$ satisfies the stationarity condition for the discrete skeleton. Conversely, if $\boldsymbol{\psi}$ is stationary for the discrete skeleton, then $\boldsymbol{\pi} = (\psi_0/\nu_0, \psi_1/\nu_1, \dots)$ satisfies the stationarity condition for the continuous-time Markov chain. If, in addition,

$$\sum_{j \in S} \frac{\psi_j}{\nu_j} = 1,$$

then $\boldsymbol{\pi}$ is a stationary distribution for the continuous-time Markov chain, but this is *never guaranteed*. However, we can renormalize: whenever the normalizing constants are finite, we can take

$$\psi_j = \frac{\pi_j \nu_j}{\sum_{k \in S} \pi_k \nu_k} \quad \text{and} \quad \pi_j = \frac{\psi_j / \nu_j}{\sum_{k \in S} \psi_k / \nu_k}$$

and in this case, we have a one-to-one correspondence between them.

The distributions $\boldsymbol{\psi}$ and $\boldsymbol{\pi}$ are not generally the same. The skeleton distribution ψ_j records the long-run proportion of visits to state j , while the continuous-time Markov chain distribution π_j records the long-run proportion of time spent in state j . Since a visit to j has mean holding time $1/\nu_j$, the continuous-time Markov chain stationary distribution weights skeleton visits by these mean holding times.

Remark 19.3

1. If the continuous-time Markov chain is irreducible and both $\boldsymbol{\pi}$ and $\boldsymbol{\psi}$ exist, then $\boldsymbol{\psi}$ is unique and hence $\boldsymbol{\pi}$ is unique.

2. It is possible for ψ to exist while π does not, because the denominator in

$$\pi_j = \frac{\psi_j/\nu_j}{\sum_{k \in S} \psi_k/\nu_k}$$

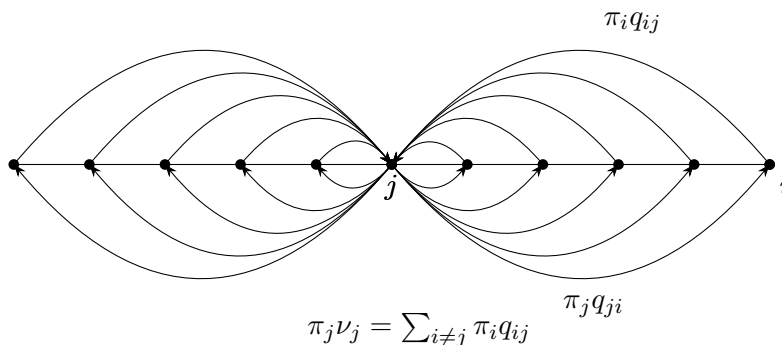
may diverge. The ν_i can be chosen arbitrarily; choosing them small enough makes normalization impossible. For example, if S is infinite, then taking $\nu_j = \psi_j$ makes $\pi = \mathbf{0}_{|S|}$, which is not a distribution.

From $\mathbf{R}^\top \pi = \mathbf{0}_{|S|}$, we obtained

$$\pi_j \nu_j = \sum_{i \neq j} \pi_i q_{ij}, \quad j \in S. \quad (19.1)$$

Here ν_j can be thought of as the rate of leaving state j , given that the chain is in state j . On the other side, $\pi_i q_{ij}$ is the rate of transitioning from state i to state j . To put it alternatively, under stationarity, $\pi_j \nu_j$ is the “probability flow” leaving state j , while $\sum_{i \neq j} \pi_i q_{ij}$ is the total “probability flow” entering state j .

Intuitively, we have that the “probability flow” leaving state j should be the same as the “probability flow” entering state j , for every $j \in S$. This is the reason why the Markov chain is stationary. Because of this interpretation, Equation (19.1) is sometimes called the **balance equation**.



Lecture 20 — Monday, November 17, 2014

The relationship between the stationary distribution ψ of the discrete skeleton and the stationary distribution π of the continuous-time Markov chain is

$$\pi_j = \frac{\psi_j/\nu_j}{\sum_{k \in S} \psi_k/\nu_k} \quad \text{and} \quad \psi_j = \frac{\pi_j \nu_j}{\sum_{k \in S} \pi_k \nu_k},$$

provided the denominators are finite. This gives two ways to find the stationary distribution.

For an irreducible continuous-time Markov chain, π_j is also the long-run proportion of time the process spends in state j , and

$$\pi_j \nu_j = \sum_{i \neq j} \pi_i q_{ij}$$

are the balance equations. Here $\pi_j \nu_j$ is the “probability flow” leaving state j , and

$$\sum_{i \neq j} \pi_i q_{ij}$$

is the “probability flow” entering state j .

Theorem 20.1 (Basic limit theorem for continuous-time Markov chains) Let $\{X(t)\}_{t \geq 0}$ be an irreducible recurrent continuous-time Markov chain. Then

$$\lim_{t \rightarrow \infty} P_{ij}(t) = \frac{\mathbb{E}[T_j]}{\mathbb{E}[R_{jj}]} = \frac{1/\nu_j}{\mathbb{E}[R_{jj}]},$$

where R_{jj} is the first return time to j . If the chain is positive recurrent, then the limiting vector is the unique stationary distribution π , so

$$\lim_{t \rightarrow \infty} P_{ij}(t) = \pi_j.$$

Remark 20.2 If the continuous-time Markov chain is positive recurrent, then $\mathbb{E}[R_{jj}] < \infty$ and $\pi_j > 0$ for all $j \in S$. If the continuous-time Markov chain is null recurrent, then $\mathbb{E}[R_{jj}] = \infty$ and no stationary distribution exists.

Partial proof. Any limiting distribution must be stationary. For any $t, s \geq 0$, the Chapman-

Kolmogorov equations gives

$$P_{ij}(t+s) = \sum_{k \in S} P_{ik}(t)P_{kj}(s).$$

If $\lim_{t \rightarrow \infty} P_{ij}(t)$ exists, then taking $t \rightarrow \infty$ yields

$$\lim_{t \rightarrow \infty} P_{ij}(t+s) = \sum_{k \in S} \left(\lim_{t \rightarrow \infty} P_{ik}(t) \right) P_{kj}(s).$$

Thus the limiting vector $\ell = (\lim_{t \rightarrow \infty} P_{i0}(t), \lim_{t \rightarrow \infty} P_{i1}(t), \dots)^\top$ satisfies $\ell = \mathbf{P}(s)^\top \ell$ for every $s \geq 0$. If $\mathbf{1}_{|S|}^\top \ell = 1$, then ℓ is stationary. $\square$

Example 20.3 Let

$$\mathbf{R} = \begin{pmatrix} -\alpha & \alpha \\ \beta & -\beta \end{pmatrix}, \quad \alpha, \beta > 0.$$

The chain is irreducible and positive recurrent. Solving $\mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_2$ and $\mathbf{1}_2^\top \boldsymbol{\pi} = 1$ gives

$$\pi_0 = \frac{\beta}{\alpha + \beta}, \quad \pi_1 = \frac{\alpha}{\alpha + \beta}.$$

Thus

$$\boldsymbol{\pi} = \left(\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta} \right)$$

is the unique stationary distribution.

The transition matrix is

$$\mathbf{P}(t) = \begin{pmatrix} \frac{\beta}{\alpha + \beta} + \frac{\alpha}{\alpha + \beta} e^{-(\alpha + \beta)t} & \frac{\alpha}{\alpha + \beta} - \frac{\alpha}{\alpha + \beta} e^{-(\alpha + \beta)t} \\ \frac{\beta}{\alpha + \beta} - \frac{\beta}{\alpha + \beta} e^{-(\alpha + \beta)t} & \frac{\alpha}{\alpha + \beta} + \frac{\beta}{\alpha + \beta} e^{-(\alpha + \beta)t} \end{pmatrix}.$$

Therefore

$$\lim_{t \rightarrow \infty} \mathbf{P}(t) = \begin{pmatrix} \frac{\beta}{\alpha + \beta} & \frac{\alpha}{\alpha + \beta} \\ \frac{\beta}{\alpha + \beta} & \frac{\alpha}{\alpha + \beta} \end{pmatrix}.$$

The limiting probability does not depend on the initial state, and each row is the unique stationary distribution.

Also, $T_0 \sim \text{Exp}(\alpha)$ and $T_1 \sim \text{Exp}(\beta)$, so

$$\mathbb{E}[R_{00}] = \mathbb{E}[T_0] + \mathbb{E}[T_1] = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}.$$

Hence

$$\frac{1/\nu_0}{\mathbb{E}[R_{00}]} = \frac{1/\alpha}{(\alpha + \beta)/(\alpha\beta)} = \frac{\beta}{\alpha + \beta} = \pi_0,$$

as predicted by the basic limit theorem.

Remark 20.4 For an irreducible continuous-time Markov chain, a stationary distribution exists if and only if the chain is positive recurrent.

For a birth and death process, recall that

$$\mathbf{R} = \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 & \cdots \\ \mu_1 & -(\lambda_1 + \mu_1) & \lambda_1 & \cdots \\ 0 & \mu_2 & -(\lambda_2 + \mu_2) & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

The balance equations $\mathbf{R}^\top \boldsymbol{\pi} = \mathbf{0}_{|S|}$ equate the rate out of each state with the rate into that state:

$$\pi_0 \lambda_0 = \pi_1 \mu_1 \quad \text{and} \quad \pi_1 (\mu_1 + \lambda_1) = \pi_0 \lambda_0 + \pi_2 \mu_2.$$

Using the first equation in the second gives

$$\pi_1 \lambda_1 = \pi_2 \mu_2.$$

Continuing inductively,

$$\pi_n \lambda_n = \pi_{n+1} \mu_{n+1} \quad \text{and hence} \quad \pi_n = \pi_0 \prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}}.$$

Normalizing gives

$$\pi_0 = \left(1 + \sum_{n=1}^{\infty} \prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}} \right)^{-1},$$

provided the sum is finite. Hence

$$\pi_n = \left(1 + \sum_{n=1}^{\infty} \prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}} \right)^{-1} \prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}}.$$

Thus a stationary distribution exists if and only if

$$\sum_{n=0}^{\infty} \prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}} < \infty,$$

in which case the stationary distribution and the limiting distribution coincide. On the other hand, if the normalizing constant is infinite, then each $\pi_n = 0$, and the Markov chain will have no stationary distribution; therefore, the chain will be either null recurrent or transient. We conclude the following:

Theorem 20.5 A birth and death process is positive recurrent if and only if

$$\sum_{n=0}^{\infty} \prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}} < \infty.$$

Lecture 21 — Wednesday, November 19, 2014

The stationary distribution π exists if and only if the Markov chain has at least one positive recurrent class, and it is unique if and only if the Markov chain has *exactly* one positive recurrent class.

Question 21.1 If the condition

$$\sum_{n=0}^{\infty} \prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}} < \infty$$

is not satisfied, how can we determine whether the birth and death process is null recurrent or transient?

Consider an irreducible birth and death process, so that $\lambda_j, \mu_j > 0$ for the relevant states. Its

discrete skeleton has transition matrix

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & \cdots \\ \frac{\mu_1}{\lambda_1 + \mu_1} & 0 & \frac{\lambda_1}{\lambda_1 + \mu_1} & \cdots \\ 0 & \frac{\mu_2}{\lambda_2 + \mu_2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Recall that

$$f_{n0} = \mathbb{P}(\text{the chain ever visits state } 0 \mid X_0 = n).$$

Then $f_{00} = 1$ if and only if the irreducible chain is recurrent, while $f_{00} < 1$ if and only if it is transient.

For a first-step analysis, we have

$$\begin{aligned} f_{00} &= f_{10}, \\ f_{10} &= \frac{\mu_1}{\lambda_1 + \mu_1} + \frac{\lambda_1}{\lambda_1 + \mu_1} f_{20}. \end{aligned}$$

Hence

$$(\lambda_1 + \mu_1)f_{10} = \mu_1 + \lambda_1 f_{20},$$

so

$$\lambda_1(f_{20} - f_{10}) = \mu_1(f_{10} - 1), \quad f_{20} - f_{10} = \frac{\mu_1}{\lambda_1}(f_{10} - 1).$$

In general, for $n \geq 1$,

$$f_{n0} = \frac{\mu_n}{\lambda_n + \mu_n} f_{n-1,0} + \frac{\lambda_n}{\lambda_n + \mu_n} f_{n+1,0},$$

which is equivalent to

$$\lambda_n(f_{n+1,0} - f_{n0}) = \mu_n(f_{n0} - f_{n-1,0}).$$

Define

$$\Delta_n := f_{n+1,0} - f_{n0}.$$

Then

$$\Delta_n = \frac{\mu_n}{\lambda_n} \Delta_{n-1}, \quad \Delta_0 = f_{10} - 1.$$

By iteration,

$$\Delta_n = \left(\prod_{i=1}^n \frac{\mu_i}{\lambda_i} \right) (f_{10} - 1),$$

that is,

$$f_{n0} - f_{n-1,0} = \left(\prod_{i=1}^n \frac{\mu_i}{\lambda_i} \right) (f_{10} - 1), \quad n \geq 1.$$

Now telescope:

$$f_{n+1,0} = f_{10} + \sum_{k=1}^n (f_{k+1,0} - f_{k0}) = f_{10} + (f_{10} - 1) \sum_{k=1}^n \prod_{i=1}^k \frac{\mu_i}{\lambda_i}.$$

Since probabilities are nonnegative, $f_{n+1,0} \geq 0$ for every n . If the chain is transient, then $f_{00} < 1$, and because $f_{00} = f_{10}$ we have $f_{10} - 1 < 0$. Therefore

$$\sum_{k=1}^n \prod_{i=1}^k \frac{\mu_i}{\lambda_i} \leq \frac{f_{10}}{1 - f_{10}} = \frac{f_{00}}{1 - f_{00}} < \infty.$$

Taking $n \rightarrow \infty$ yields the necessary condition that the series

$$\sum_{k=1}^{\infty} \frac{\mu_k \cdots \mu_1}{\lambda_k \cdots \lambda_1} = \sum_{k=1}^{\infty} \prod_{i=1}^k \frac{\mu_i}{\lambda_i}$$

converges. Conversely, one (not us) can show that if this series converges, then $f_{00} < 1$, so the chain is transient. Hence the chain is transient if and only if

$$\sum_{k=1}^{\infty} \prod_{i=1}^k \frac{\mu_i}{\lambda_i} < \infty.$$

If the positive recurrence condition fails and this transience condition also fails, then the chain is null recurrent.

Example 21.2 For the $M/M/1$ queue,

$$\lambda_n = \lambda \quad (n = 0, 1, 2, \dots), \quad \mu_n = \mu \quad (n = 1, 2, \dots).$$

The transience criterion becomes

$$\sum_{k=1}^{\infty} \left(\frac{\mu}{\lambda} \right)^k < \infty,$$

which holds if and only if $\mu < \lambda$. Meanwhile, the positive recurrence criterion gives positive recurrence if $\lambda < \mu$. Therefore the $M/M/1$ queue is positive recurrent if $\lambda < \mu$, transient if $\lambda > \mu$, and null recurrent if $\lambda = \mu$.

Continuous Phase-Type Distributions

Let $\{X(t)\}_{t \geq 0}$ be a continuous-time Markov chain with transient states

$$E = \{1, 2, \dots, m\}$$

and a single absorbing state 0. With the absorbing state listed first, the generator has block form

$$\mathbf{R} = \begin{pmatrix} 0 & \mathbf{0}_{1 \times m} \\ \mathbf{t}^0 & \mathbf{T} \end{pmatrix},$$

where $\mathbf{T}$ is the $m \times m$ transient block and $\mathbf{t}^0$ is the column vector of rates from the transient states to the absorbing state. Since the rows of $\mathbf{R}$ sum to zero,

$$\mathbf{t}^0 + \mathbf{T}\mathbf{1}_m = 0 \implies \mathbf{t}^0 = -\mathbf{T}\mathbf{1}_m.$$

Let $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_m)^\top$ be the initial probability column vector over the transient states. Then

$$\alpha_i = \mathbb{P}(X(0) = i) \quad \text{and} \quad \alpha_0 = \mathbb{P}(X(0) = 0) = 1 - \sum_{i=1}^m \alpha_i = 1 - \boldsymbol{\alpha}^\top \mathbf{1}_m.$$

Define the absorption time

$$Y = \min\{t \geq 0 : X(t) = 0\}.$$

For $y \geq 0$,

$$\begin{aligned} \mathbb{P}(Y > y) &= \mathbb{P}(X(y) \in E) \\ &= \sum_{i=1}^m \alpha_i \mathbb{P}(X(y) \in E \mid X(0) = i) \\ &= \sum_{i=1}^m \alpha_i \sum_{j=1}^m \mathbb{P}(X(y) = j \mid X(0) = i) \\ &= \sum_{i=1}^m \alpha_i \sum_{j=1}^m P_{ij}(y). \end{aligned}$$

The transition matrix at time y is

$$\mathbf{P}(y) = e^{\mathbf{R}y} = \mathbf{I} + \sum_{n=1}^{\infty} \frac{y^n}{n!} \mathbf{R}^n.$$

Using the block structure of $\mathbf{R}$,

$$\mathbf{R}^n = \begin{pmatrix} 0 & \mathbf{0}_{1 \times m} \\ ?_n & \mathbf{T}^n \end{pmatrix},$$

where $?_n$ is something we don't care about. Therefore,

$$\mathbf{P}(y) = \mathbf{I} + \sum_{n=1}^{\infty} \frac{y^n}{n!} \begin{pmatrix} 0 & \mathbf{0}_{1 \times m} \\ ?_n & \mathbf{T}^n \end{pmatrix} = \begin{pmatrix} 1 & \mathbf{0}_{1 \times m} \\ ?? & e^{\mathbf{T}y} \end{pmatrix}$$

So the transient-to-transient block of $\mathbf{P}(y)$ is $e^{\mathbf{T}y}$, and because the rows sum to 1, we see that the full matrix has the form

$$\mathbf{P}(y) = \begin{pmatrix} 1 & \mathbf{0}_{1 \times m} \\ \mathbf{1}_m - e^{\mathbf{T}y} \mathbf{1}_m & e^{\mathbf{T}y} \end{pmatrix}.$$

Consequently,

$$\mathbb{P}(Y > y) = \sum_{i=1}^m \alpha_i \sum_{j=1}^m (e^{\mathbf{T}y})_{ij} = \boldsymbol{\alpha}^\top e^{\mathbf{T}y} \mathbf{1}_m$$

and so the cumulative distribution function of Y is

$$F_Y(y) = \mathbb{P}(Y \leq y) = 1 - \boldsymbol{\alpha}^\top e^{\mathbf{T}y} \mathbf{1}_m.$$

In particular,

$$\mathbb{P}(Y = 0) = F_Y(0) = 1 - \boldsymbol{\alpha}^\top \mathbf{1}_m = \alpha_0.$$

For $y > 0$, the density is

$$f_Y(y) = \frac{d}{dy} F_Y(y) = -\boldsymbol{\alpha}^\top \mathbf{T} e^{\mathbf{T}y} \mathbf{1}_m = \boldsymbol{\alpha}^\top e^{\mathbf{T}y} \mathbf{t}^0.$$

Thus Y has a point mass α_0 at 0 and density $\alpha e^{\mathbf{T}y} \mathbf{t}^0$ for $y > 0$.

Lecture 22 — Monday, November 24, 2014

Definition 22.1 A random variable Y constructed as the absorption time above is said to have a **continuous phase-type distribution** with representation $(\boldsymbol{\alpha}, \mathbf{T})$. We write

$$Y \sim \text{CPH}_m(\boldsymbol{\alpha}, \mathbf{T}).$$

Here α is the initial distribution over the m transient phases, and $\mathbf{T}$ is the transient part of the generator matrix.

Example 22.2 The exponential distribution is a continuous phase-type distribution. If $Y \sim \text{Exp}(\lambda)$, take one transient state and one absorbing state, with

$$\mathbf{R} = \begin{pmatrix} 0 & 0 \\ \lambda & -\lambda \end{pmatrix}, \quad \alpha = (1), \quad T = (-\lambda), \quad \text{and} \quad t^0 = (\lambda).$$

Then $Y \sim \text{CPH}_1(1, -\lambda)$.

We list out some properties of continuous phase-type distributions.

1. *Independent sums.* If $X \sim \text{CPH}_m(\alpha, \mathbf{T})$ and $Y \sim \text{CPH}_n(\beta, \mathbf{S})$ are independent, then

$$X + Y \sim \text{CPH}_{m+n}(\delta, \mathbf{D}),$$

where

$$\delta = (\alpha, \alpha_0 \beta) \quad \text{and} \quad \mathbf{D} = \begin{pmatrix} \mathbf{T} & t^0 \beta \\ 0 & \mathbf{S} \end{pmatrix}.$$

Thus continuous phase-type distributions are closed under independent sums.

As a special case, the **Erlang distribution** with density

$$f(x) = \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!}, \quad x > 0,$$

is continuous phase-type, since an $\text{Erlang}(n, \lambda)$ random variable is the sum of n independent $\text{Exp}(\lambda)$ random variables. One representation uses

$$\alpha = (1, 0, \dots, 0), \quad \mathbf{T} = \begin{pmatrix} -\lambda & \lambda & 0 & \dots \\ 0 & -\lambda & \lambda & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

The same construction works for independent sums of exponential random variables with different parameters.

2. *Finite mixtures.* If $X \sim \text{CPH}_m(\alpha, \mathbf{T})$ and $Y \sim \text{CPH}_n(\beta, \mathbf{S})$, and

$$Z = \begin{cases} X, & \text{with probability } p, \\ Y, & \text{with probability } 1 - p, \end{cases}$$

where the choice is independent of X and Y . Then $F_Z = pF_X + (1-p)F_Y$ and $Z \sim \text{CPH}_{m+n}(\gamma, \mathbf{G})$, where

$$\gamma = \begin{pmatrix} p\boldsymbol{\alpha} \\ (1-p)\boldsymbol{\beta} \end{pmatrix} \quad \text{and} \quad \mathbf{G} = \begin{pmatrix} \mathbf{T} & \mathbf{0}_{m \times n} \\ \mathbf{0}_{n \times m} & \mathbf{S} \end{pmatrix}.$$

Thus mixtures of exponential or Erlang distributions are again continuous phase-type. For example,

$$f(x) = p \frac{\lambda^2 x e^{-\lambda x}}{1!} + (1-p) \frac{\lambda^7 x^6 e^{-\lambda x}}{6!}, \quad x > 0,$$

is a mixture of $\text{Erlang}(2, \lambda)$ and $\text{Erlang}(7, \lambda)$ distributions, hence is continuous phase-type.

A direct block construction gives a nine-phase representation:

$$Z \sim \text{CPH}_9(\tilde{\boldsymbol{\alpha}}, \tilde{\mathbf{T}}), \quad \tilde{\boldsymbol{\alpha}} = \begin{pmatrix} p \\ 0 \\ 1-p \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

with

$$\tilde{\mathbf{T}} = \begin{pmatrix} \mathbf{T}_2 & \mathbf{0}_{2 \times 7} \\ \mathbf{0}_{7 \times 2} & \mathbf{T}_7 \end{pmatrix},$$

where $\mathbf{T}_r$ is the $r \times r$ Erlang transient block with $-\lambda$ on the diagonal and λ on the super-diagonal.

We can also obtain a smaller representation. Let

$$\mathbf{T}_7 = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & 0 & 0 & 0 \\ 0 & -\lambda & \lambda & 0 & 0 & 0 & 0 \\ 0 & 0 & -\lambda & \lambda & 0 & 0 & 0 \\ 0 & 0 & 0 & -\lambda & \lambda & 0 & 0 \\ 0 & 0 & 0 & 0 & -\lambda & \lambda & 0 \\ 0 & 0 & 0 & 0 & 0 & -\lambda & \lambda \\ 0 & 0 & 0 & 0 & 0 & 0 & -\lambda \end{pmatrix}, \quad \boldsymbol{\beta} = (1-p, 0, 0, 0, 0, p, 0).$$

Starting in phase 1 gives $\text{Erlang}(7, \lambda)$, while starting in phase 6 gives $\text{Erlang}(2, \lambda)$. Therefore

$$Z \sim \text{CPH}_7(\boldsymbol{\beta}, \mathbf{T}_7).$$

This shows explicitly that continuous phase-type representations are not unique.

3. *Finite moments.* If $X \sim \text{CPH}_m(\alpha, \mathbf{T})$, then X has all finite moments. Let

$$\mathcal{L}(s) = \mathbb{E}[e^{-sX}]$$

be its Laplace transform. Since X has point mass α_0 at 0 and density $\boldsymbol{\alpha}^\top e^{\mathbf{T}x} \mathbf{t}^0$ for $x > 0$,

$$\begin{aligned} \mathcal{L}(s) &= \alpha_0 + \int_0^\infty e^{-sx} \boldsymbol{\alpha}^\top e^{\mathbf{T}x} \mathbf{t}^0 \, dx \\ &= \alpha_0 + \boldsymbol{\alpha}^\top \int_0^\infty e^{(\mathbf{T}-s\mathbf{I})x} \, dx \mathbf{t}^0 \\ &= \alpha_0 + \boldsymbol{\alpha}^\top \left[e^{(\mathbf{T}-s\mathbf{I})x} (\mathbf{T}-s\mathbf{I})^{-1} \right]_{x=0}^\infty \mathbf{t}^0 \\ &= \alpha_0 + \boldsymbol{\alpha}^\top (s\mathbf{I} - \mathbf{T})^{-1} \mathbf{t}^0 \end{aligned}$$

for $s \geq 0$. The boundary term at ∞ is 0 because all eigenvalues of $\mathbf{T} - s\mathbf{I}$ have strictly negative real part for $s \geq 0$.

To recover moments, note that for each $n \geq 1$,

$$\frac{d^n}{ds^n} \mathcal{L}(s) = \frac{d^n}{ds^n} \mathbb{E}[e^{-sX}] = \mathbb{E} \left[\frac{d^n}{ds^n} e^{-sX} \right] = (-1)^n \mathbb{E}[X^n e^{-sX}],$$

so at $s = 0$,

$$\mathbb{E}[X^n] = (-1)^n \left. \frac{d^n}{ds^n} \mathcal{L}(s) \right|_{s=0}.$$

Now differentiate the matrix-inverse term. Since

$$\frac{d}{ds} (s\mathbf{I} - \mathbf{T})^{-1} = -(s\mathbf{I} - \mathbf{T})^{-2},$$

induction gives

$$\frac{d^n}{ds^n} (s\mathbf{I} - \mathbf{T})^{-1} = (-1)^n n! (s\mathbf{I} - \mathbf{T})^{-n-1}.$$

Therefore

$$\frac{d^n}{ds^n} \mathcal{L}(s) = (-1)^n n! \boldsymbol{\alpha}^\top (s\mathbf{I} - \mathbf{T})^{-n-1} \mathbf{t}^0.$$

Using $\mathbf{t}^0 = -\mathbf{T}\mathbf{1}_m$ and evaluating at $s = 0$,

$$\begin{aligned} \mathbb{E}[X^n] &= (-1)^n \left. \frac{d^n}{ds^n} \mathcal{L}(s) \right|_{s=0} \\ &= n! \boldsymbol{\alpha}^\top (-\mathbf{T})^{-n-1} (-\mathbf{T}) \mathbf{1}_m \\ &= n! \boldsymbol{\alpha}^\top (-\mathbf{T})^{-n} \mathbf{1}_m, \quad n = 1, 2, 3, \dots \end{aligned}$$

Hence every moment is finite. In particular,

$$\mathbb{E}[X] = \boldsymbol{\alpha}^\top (-\mathbf{T})^{-1} \mathbf{1}_m.$$

Remark 22.3 Continuous phase-type distributions are dense in a broad class of continuous distributions; that is, many continuous distributions can be approximated arbitrarily closely by continuous phase-type distributions.

Lecture 23 — Wednesday, November 26, 2014

4. Martingales

Definitions and Examples

Definition 23.1 A stochastic process $\{X_n\}_{n \geq 0}$ is a **martingale** if

1. $\mathbb{E}[|X_n|] < \infty$ for all n ;
2. $\mathbb{E}[X_{n+1} \mid X_0, \dots, X_n] = X_n$ for all n .

The guiding intuition is that a martingale models a fair game: conditional on the present and the past, the expected future value equals the current value.

Example 23.2 Let $S_0 = 0$ and let $X_1, X_2, \dots$ be independent random variables with

$$\mathbb{P}(X_i = 1) = \frac{1}{2} = \mathbb{P}(X_i = -1).$$

The symmetric random walk $\{S_n\}_{n \geq 0}$ is defined by

$$S_n = \sum_{i=1}^n X_i.$$

Then $\{S_n\}_{n \geq 0}$ is a martingale. Indeed, $|S_n| \leq n$, so $\mathbb{E}[|S_n|] < \infty$, and

$$\mathbb{E}[S_{n+1} \mid S_0, \dots, S_n] = \mathbb{E}[S_n + X_{n+1} \mid S_0, \dots, S_n] = S_n + \mathbb{E}[X_{n+1}] = S_n.$$

Example 23.3 Let $X_1, X_2, \dots$ be independent random variables with $\mathbb{E}[X_i] = 1$ and $\mathbb{E}[|X_i|] < \infty$. Define

$$Z_n = \prod_{i=1}^n X_i.$$

Assuming $\mathbb{E}[|Z_n|] < \infty$ for each n , the process $\{Z_n\}_{n \geq 0}$ is a martingale, since

$$\mathbb{E}[Z_{n+1} \mid Z_0, \dots, Z_n] = \mathbb{E}[Z_n X_{n+1} \mid Z_0, \dots, Z_n] = Z_n \mathbb{E}[X_{n+1}] = Z_n.$$

Proposition 23.4 If $\{X_n\}_{n \geq 0}$ is a martingale, then $\mathbb{E}[X_n] = \mathbb{E}[X_0]$ for every n .

Proof. By iterating conditional expectation,

$$\mathbb{E}[X_n] = \mathbb{E}[\mathbb{E}[X_n \mid X_0, \dots, X_{n-1}]] = \mathbb{E}[X_{n-1}] = \dots = \mathbb{E}[X_0].$$

□

Proposition 23.5 If $\{X_n\}_{n \geq 0}$ is a martingale with finite variances, then

$$\text{Var}(X_{n+1}) \geq \text{Var}(X_n)$$

for every n .

Proof. Let $Y = X_{n+1} - X_n$. Since

$$\mathbb{E}[Y \mid X_0, \dots, X_n] = 0,$$

we have

$$\text{Cov}(Y, X_n) = \mathbb{E}[Y X_n] - \mathbb{E}[Y] \mathbb{E}[X_n] = \mathbb{E}[\mathbb{E}[Y X_n \mid X_0, \dots, X_n]] = \mathbb{E}[X_n \mathbb{E}[Y \mid X_0, \dots, X_n]] = 0.$$

Therefore

$$\text{Var}(X_{n+1}) = \text{Var}(X_n + Y) = \text{Var}(X_n) + \text{Var}(Y) \geq \text{Var}(X_n).$$

□

Stopping Times and Optional Sampling

For deterministic times n , a martingale satisfies $\mathbb{E}[X_n] = \mathbb{E}[X_0]$. For random times T , this conclusion can fail.

Example 23.6 Let $\{S_n\}$ be the simple symmetric random walk and define

$$T = \min \left\{ 0 \leq n \leq N : S_n = \max_{0 \leq i \leq N} S_i \right\}.$$

This is the first time the walk reaches its maximum over the full interval $\{0, \dots, N\}$. Then $S_T \geq 0$ and $\mathbb{P}(S_T > 0) > 0$, so

$$\mathbb{E}[S_T] > 0 = \mathbb{E}[S_0].$$

The issue is that T depends on future information.

Definition 23.7 A non-negative integer-valued random variable T is a **stopping time** for the stochastic process $\{X_n\}_{n \geq 0}$ if the event $\{T = n\}$ is determined by $X_0, \dots, X_n$ for every n . Equivalently, for each n there is a function f_n such that

$$\mathbb{1}_{\{T=n\}} = f_n(X_0, \dots, X_n).$$

Intuitively, if T is a stopping time, then we know that the process has arrived when it arrives. That clears up everything.

Example 23.8 The first hitting time of a level h ,

$$T_h = \min\{n : X_n = h\},$$

is a stopping time. By contrast, the last time the process hits a level h before time N ,

$$T_h^e = \max\{n \leq N : X_n = h\},$$

is generally not a stopping time, since deciding whether a time is the last visit requires future information.

Stopping times alone are not enough to say very much. For the simple symmetric random walk,

the hitting time

$$T_1 = \min\{n : S_n = 1\}$$

is a stopping time, but

$$\mathbb{E}[S_{T_1}] = 1 \neq 0 = \mathbb{E}[S_0].$$

Additional hypotheses are needed.

Theorem 23.9 (Optional sampling) Let $\{X_n\}_{n \geq 0}$ be a martingale, and let T be a stopping time with $\mathbb{P}(T < \infty) = 1$. If

$$\mathbb{E}[|X_T|] < \infty$$

and

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n| \mathbb{1}_{\{T > n\}}] = 0,$$

then

$$\mathbb{E}[X_T] = \mathbb{E}[X_0].$$

In particular, the conclusion holds when T is bounded or when T is **uniformly integrable**:

$$\lim_{K \rightarrow \infty} \sup_n \mathbb{E}[|X_n| \mathbb{1}_{\{|X_n| > K\}}] = 0.$$

Example 23.10 (Gambler's ruin: symmetric case) Suppose $X_0 = m$, and from each interior state the gambler moves up or down by one with probabilities $p = q = 1/2$. The game ends when the wealth reaches 0 or N . Let

$$T = \min\{n : X_n = 0 \text{ or } X_n = N\}.$$

Then $\{X_n\}$ is a martingale, X_T is bounded, and the optional sampling theorem gives

$$\mathbb{E}[X_T] = \mathbb{E}[X_0] = m.$$

Since

$$\mathbb{E}[X_T] = 0 \cdot \mathbb{P}(X_T = 0) + N \cdot \mathbb{P}(X_T = N),$$

we obtain

$$\mathbb{P}(X_T = N) = \frac{m}{N}, \quad \mathbb{P}(X_T = 0) = 1 - \frac{m}{N}.$$

Lecture 24 — Monday, December 01, 2014

We continue our foray into martingales.

Example 24.1 (Gambler's ruin: asymmetric case) Now suppose $p \neq q$, where $q = 1 - p$. Put

$$r = \frac{q}{p}, \quad Y_n = r^{X_n}.$$

Then $\{Y_n\}$ is a martingale, since, conditional on $X_n = x$,

$$\mathbb{E}[Y_{n+1} \mid X_n = x] = pr^{x+1} + qr^{x-1} = r^x \left(pr + \frac{q}{r} \right) = r^x(q + p) = r^x.$$

Alternatively, write $Y_n = Y_{n-1}Z_n$, where

$$Z_n = \begin{cases} q/p & \text{if } X_n = X_{n-1} + 1 \\ p/q & \text{if } X_n = X_{n-1} - 1 \end{cases}$$

and the Z_n are all independent with $\mathbb{E}[Z_n] = (q/p)p + (p/q)q = 1$. By [Example 23.3](#),

$$Y_n = Y_0 \prod_{i=1}^n Z_i = \left(\frac{q}{p} \right)^m \prod_{i=1}^n Z_i$$

is a martingale.

With the same stopping time

$$T = \min\{n : X_n = 0 \text{ or } X_n = N\},$$

the stopped value Y_T is bounded. The optional sampling theorem gives

$$\mathbb{E}[Y_T] = \mathbb{E}[Y_0] = r^m.$$

If $a = \mathbb{P}(X_T = 0)$ is the probability of ruin, then

$$r^m = \mathbb{E}[r^{X_T}] = a + r^N(1 - a).$$

Solving for a gives

$$\mathbb{P}(X_T = 0) = \frac{r^N - r^m}{r^N - 1} = \frac{(q/p)^N - (q/p)^m}{(q/p)^N - 1}.$$

This is substantially easier than the old-school way of deriving the ruin probability!

Martingale Convergence

Theorem 24.2 (Martingale convergence theorem) Let $\{X_n\}_{n \geq 0}$ be a martingale. If there exists $M < \infty$ such that

$$\sup_{n \geq 1} \mathbb{E}[|X_n|] \leq M$$

for all n , then $\lim_{n \rightarrow \infty} X_n$ exists and is finite with probability 1.

The condition $\sup_{n \geq 1} \mathbb{E}[|X_n|] \leq M < \infty$ for all n means that $\{X_n\}$ is uniformly bounded with respect to the 1-norm.

The proof is omitted since it uses measure theory, which we are not equipped with.

Corollary 24.3 If $\{X_n\}_{n \geq 0}$ is a non-negative martingale, then $\lim_{n \rightarrow \infty} X_n$ exists and is finite with probability 1.

Proof. Since $X_n \geq 0$,

$$\mathbb{E}[|X_n|] = \mathbb{E}[X_n] = \mathbb{E}[X_0]$$

for every n . Thus the martingale is uniformly bounded in L^1 , and the martingale convergence theorem applies. $\square$

Example 24.4 Consider a discrete stock price model with $X_0 = a$ and

$$X_n = a \prod_{i=1}^n Z_i, \quad \mathbb{P}(Z_i = 1.2) = \mathbb{P}(Z_i = 0.8) = \frac{1}{2},$$

where $Z_1, Z_2, \dots$ are independent. Each step is an up-move or down-move of 20%.

First, $\{X_n\}$ is a positive martingale:

$$\mathbb{E}[X_{n+1} \mid X_0, \dots, X_n] = \mathbb{E}[X_n Z_{n+1} \mid X_0, \dots, X_n] = X_n \mathbb{E}[Z_{n+1}] = X_n,$$

since $\mathbb{E}[Z_{n+1}] = \frac{1}{2}(1.2) + \frac{1}{2}(0.8) = 1$. Therefore $X_n \geq 0$ and the martingale convergence theorem implies that $X_n \rightarrow X_\infty$ almost surely for some finite random variable X_∞ .

To identify the limit, write $U_n = \sum_{i=1}^n \mathbb{1}_{\{Z_i=1.2\}}$ and $D_n = n - U_n$. Then

$$X_n = a 1.2^{U_n} 0.8^{D_n} = a \exp(U_n \log(1.2) + D_n \log(0.8)).$$

Hence

$$\frac{1}{n} \log \left(\frac{X_n}{a} \right) = \frac{U_n}{n} \log(1.2) + \frac{D_n}{n} \log(0.8) \longrightarrow \frac{1}{2} \log(1.2) + \frac{1}{2} \log(0.8) = \frac{1}{2} \log(0.96) < 0.$$

Therefore $\log(X_n/a) \rightarrow -\infty$, so $X_n \rightarrow 0$ almost surely. In particular, $X_\infty = 0$ almost surely.

An intuitive way to read this is: in the long run, the number of up-moves and down-moves is approximately the same, and one up with one down multiplies the price by $1.2 \times 0.8 = 0.96 < 1$. Repeated pairs drive the price downward.

This does not contradict the martingale property. We still have $\mathbb{E}[X_n] = a$ for every n . The mean is preserved by increasingly rare trajectories on which X_n becomes very large, while with overwhelming probability the process drifts toward 0.